
Solutions Manual

**Fundamentals of
Engineering
Electromagnetics**

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PREFACE

This solutions manual is prepared for the convenience of those professors who assign my Fundamentals of Engineering Electromagnetics as the textbook for their classes. All problems in the book are solved in sufficient detail so that no trouble should be encountered in arriving at the final results[†]. To lend confidence to the students who are assigned to do the problems, answers to odd-numbered problems are given at the end of the book. I have asked my publisher, the Addison-Wesley Publishing Company, to exercise strict control in sending out this solutions manual to prevent it from getting into the hands of students.

I realize that, no matter how careful I have endeavored to be, occasional errors may still exist. I should be grateful if you would be kind enough to notify me as you discover them either in the book or in this manual.

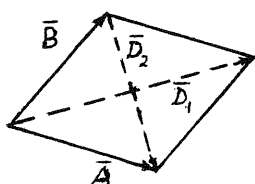
D.K.C.

[†]In this manual letters with an overbar represent vector quantities which are printed with a boldface in the book. A vector from point P_1 to point P_2 is indicated by $\overrightarrow{P_1P_2}$.

Chapter 2

Vector Analysis

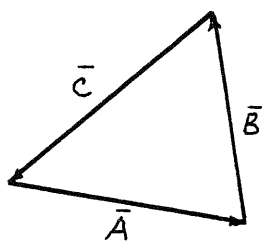
P. 2-1 Denoting the diagonals of the rhombus by $\vec{D}_1$ and $\vec{D}_2$, we have:



$$(a) \quad \vec{D}_1 = \vec{A} + \vec{B}, \\ \vec{D}_2 = \vec{A} - \vec{B}.$$

$$(b) \quad \vec{D}_1 \cdot \vec{D}_2 = (\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) \\ = \vec{A} \cdot \vec{A} - \vec{B} \cdot \vec{B} = 0, \\ \text{since } |\vec{A}| = |\vec{B}|. \\ \text{Thus, } \vec{D}_1 \perp \vec{D}_2.$$

P. 2-2



$$\vec{A} + \vec{B} + \vec{C} = 0.$$

$$\vec{A} \times : \vec{A} \times \vec{B} = \vec{C} \times \vec{A}.$$

$$\vec{C} \times : \vec{C} \times \vec{A} = \vec{B} \times \vec{C}.$$

$$\vec{B} \times : \vec{B} \times \vec{C} = \vec{A} \times \vec{B}.$$

Magnitude relations:

$$AB \sin \theta_{AB} = CA \sin \theta_{CA} = BC \sin \theta_{BC}.$$

Hence,

$$\frac{A}{\sin \theta_{BC}} = \frac{B}{\sin \theta_{CA}} = \frac{C}{\sin \theta_{AB}}. \quad (\text{Law of Sines.})$$

P. 2-3 a) $\vec{a}_B = \frac{\vec{a}_x 4 - \vec{a}_y 6 + \vec{a}_z 12}{\sqrt{4^2 + 6^2 + 12^2}} = \vec{a}_x \frac{2}{7} - \vec{a}_y \frac{3}{7} + \vec{a}_z \frac{6}{7}.$

b) $\vec{B} - \vec{A} = -\vec{a}_x 2 - \vec{a}_y 8 + \vec{a}_z 15, \quad |\vec{B} - \vec{A}| = \sqrt{2^2 + 8^2 + 15^2} = 17.1.$

c) $\vec{A} \cdot \vec{a}_B = 6 \times \frac{2}{7} - 2 \times \frac{3}{7} - 3 \times \frac{6}{7} = -17.1.$

d) $\vec{B} \cdot \vec{A} = 24 - 12 - 36 = -24.$

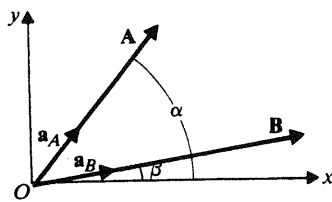
e) $\vec{B} \cdot \vec{a}_A = \frac{\vec{B} \cdot \vec{A}}{|\vec{A}|} = \frac{-24}{\sqrt{6^2 + 2^2 + 3^2}} = -\frac{24}{7} = -3.43.$

f) $\cos \theta_{AB} = \frac{\vec{B} \cdot \vec{A}}{BA} = \frac{-24}{14 \times 7} = -0.245, \quad \theta_{AB} = 180^\circ - 75.8^\circ = 104.2^\circ$

$$9) \quad \bar{A} \times \bar{C} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ 6 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = -\bar{a}_x 4 - \bar{a}_y 3 - \bar{a}_z 10$$

$$11) \quad \bar{A} \cdot (\bar{B} \times \bar{C}) = (\bar{A} \times \bar{B}) \cdot \bar{C} = -(\bar{A} \times \bar{C}) \cdot \bar{B} = -[(-4)4 + (-3)(-6) + (-10)12] = -118.$$

P. 2-4



$$\begin{aligned} \bar{a}_A &= \bar{a}_x \cos \alpha + \bar{a}_y \sin \alpha, \\ \bar{a}_B &= \bar{a}_x \cos \beta + \bar{a}_y \sin \beta. \end{aligned}$$

$$a) \quad \bar{a}_A \cdot \bar{a}_B = \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

$$\begin{aligned} b) \quad \bar{a}_B \times \bar{a}_A &= \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \cos \beta & \sin \beta & 0 \\ \cos \alpha & \sin \alpha & 0 \end{vmatrix} = \bar{a}_z (\sin \alpha \cos \beta - \cos \alpha \sin \beta) \\ &= \bar{a}_z \sin(\alpha - \beta). \end{aligned}$$

$$\therefore \sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

$$\begin{aligned} \text{P. 2-5 } a) \quad \vec{P_1 P_2} &= \vec{OP_2} - \vec{OP_1} = -\bar{a}_x 4 + \bar{a}_y + \bar{a}_z 3, \\ \vec{P_2 P_3} &= \vec{OP_3} - \vec{OP_2} = \bar{a}_x 6 - \bar{a}_y 5 + \bar{a}_z, \\ \vec{P_1 P_3} &= \vec{OP_3} - \vec{OP_1} = \bar{a}_x 2 - \bar{a}_y 4 + \bar{a}_z 4. \end{aligned}$$

$$\vec{P_1 P_2} \cdot \vec{P_1 P_3} = 0. \rightarrow \text{Right angle at corner } P_1.$$

$$b) \quad \text{Area of triangle} = \frac{1}{2} |\vec{P_1 P_2} \times \vec{P_1 P_3}| = \frac{1}{2} |\vec{P_1 P_2}| |\vec{P_1 P_3}| = 15.3.$$

$$\text{P. 2-6 } a) \quad \vec{P_1 P_2} = \bar{a}_x 2 + \bar{a}_y 4 - \bar{a}_z 4, \quad |\vec{P_1 P_2}| = \sqrt{2^2 + 4^2 + 4^2} = 6.$$

b) Perpendicular distance from P_3 to the line

$$\begin{aligned} &= |\vec{P_3 P_1} \times \bar{a}_{P_1 P_2}| = |(\vec{OP_1} - \vec{OP_3}) \times \frac{1}{6} \vec{P_1 P_2}| \\ &= |(-\bar{a}_x 5 - \bar{a}_y) \times \frac{1}{6} (\bar{a}_x 2 + \bar{a}_y 4 - \bar{a}_z 4)| = \frac{1}{6} |\bar{a}_x 4 - \bar{a}_y 20 - \bar{a}_z 18| = 4.53. \end{aligned}$$

P.2-7 Given: $\bar{A} = \bar{a}_x 5 - \bar{a}_y 2 + \bar{a}_z$.

a) Let $\bar{a}_B = \bar{a}_x B_x + \bar{a}_y B_y + \bar{a}_z B_z$,
where $(B_x^2 + B_y^2 + B_z^2)^{1/2} = 1$. (1)

$$\bar{a}_B \parallel \bar{A} \text{ requires } \bar{a}_B \times \bar{A} = 0 = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ B_x & B_y & B_z \\ 5 & -2 & 1 \end{vmatrix},$$

where yields: $B_y + 2B_z = 0$, (2a)

$$-B_x + 5B_z = 0, \quad \text{(2b)}$$

$$-2B_x - 5B_y = 0. \quad \text{(2c)}$$

Equations (2a), (2b), and (2c) are not all independent:
Solving Eqs. (1) and (2), we obtain

$$B_x = \frac{5}{\sqrt{30}}, \quad B_y = -\frac{2}{\sqrt{30}}, \quad \text{and } B_z = \frac{1}{\sqrt{30}}$$

$$\therefore \bar{a}_B = \frac{1}{\sqrt{30}} (\bar{a}_x 5 - \bar{a}_y 2 + \bar{a}_z).$$

b) Let $\bar{a}_C = \bar{a}_x C_x + \bar{a}_y C_y + \bar{a}_z C_z$, where $C_z = 0$,
and $C_x^2 + C_y^2 = 1$. (3)

$$\bar{a}_C \perp \bar{A} \text{ requires } \bar{a}_C \cdot \bar{A} = 0, \text{ or}$$

$$5C_x - 2C_y = 0. \quad \text{(4)}$$

Solution of Eqs. (3) and (4) yields

$$C_x = \frac{2}{\sqrt{29}}, \quad \text{and } C_y = \frac{5}{\sqrt{29}}.$$

$$\therefore \bar{a}_C = \frac{1}{\sqrt{29}} (\bar{a}_x 2 + \bar{a}_y 5).$$

P.2-8 Given: $\bar{A} = \bar{A}_1 + \bar{A}_2 = \bar{a}_x 2 - \bar{a}_y 5 + \bar{a}_z 3$,

$$\bar{B} = -\bar{a}_x + \bar{a}_y 4,$$

$$\bar{A}_1 \perp \bar{B} \longrightarrow \bar{A}_1 \cdot \bar{B} = 0,$$

$$\bar{A}_2 \parallel \bar{B} \longrightarrow \bar{A}_2 \times \bar{B} = 0.$$

Solving, we have

$$\bar{A}_1 = \frac{3}{17} (\bar{a}_x 4 + \bar{a}_y + \bar{a}_z 17) \text{ and } \bar{A}_2 = \frac{22}{17} (\bar{a}_x - \bar{a}_y 4).$$

P. 2-10 $\vec{OP}_1 = -\bar{a}_x - \bar{a}_z$,
 $\vec{OP}_2 = \bar{a}_x(r \cos \phi) + \bar{a}_y(r \sin \phi) + \bar{a}_z z$
 $= \bar{a}_x(-\frac{3}{2}) + \bar{a}_y \frac{\sqrt{3}}{2} + \bar{a}_z$,
 $\vec{P_1 P_2} = \vec{OP}_2 - \vec{OP}_1 = -\bar{a}_x \frac{1}{2} + \bar{a}_y \frac{\sqrt{3}}{2} + \bar{a}_z 3$, $|\vec{P_1 P_2}| = \sqrt{10}$.
 At $P_1(-1, 0, -2)$, $\vec{A}_{P_1} = -\bar{a}_x 2 + \bar{a}_z$.
 $\vec{A}_{P_1} \cdot \vec{a}_{P_1 P_2} = \vec{A}_{P_1} \cdot \frac{\vec{P_1 P_2}}{|\vec{P_1 P_2}|} = \frac{4}{\sqrt{10}} = 1.265$

P. 2-11 a) $x = r \cos \phi = 3 \cos 240^\circ = -\frac{3}{2}$,
 $y = r \sin \phi = 3 \sin 240^\circ = -3\sqrt{3}/2$,
 $z = -4$ } $(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}, -4)$
 b) $R = (r^2 + z^2)^{1/2} = (3^2 + 4^2)^{1/2} = 5$,
 $\theta = \tan^{-1}(r/z) = \tan^{-1}(\frac{3}{-4}) = 143.1^\circ$,
 $\phi = 4\pi/3 = 240^\circ$ } $(5, 143.1^\circ, 240^\circ)$

P. 2-12 a) $-\sin \phi$, b) $\sin \theta \sin \phi$, c) $\cos \theta$,
 d) $-\bar{a}_z \cos \phi$, e) $-\bar{a}_\phi \cos \theta$, f) $-\bar{a}_\phi \cos \theta$.

P. 2-13 a) In Cartesian coordinates, $\vec{A} = \bar{a}_x A_x + \bar{a}_y A_y + \bar{a}_z A_z$.
 $A_r = \vec{a}_r \cdot \vec{A} = (\vec{a}_r \cdot \bar{a}_x) A_x + (\vec{a}_r \cdot \bar{a}_y) A_y + (\vec{a}_r \cdot \bar{a}_z) A_z$
 $= A_x \cos \phi + A_y \sin \phi$
 b) In spherical coordinates, $\vec{A} = \bar{a}_R A_R + \bar{a}_\theta A_\theta + \bar{a}_\phi A_\phi$.
 $A_r = \vec{a}_r \cdot \vec{A} = (\vec{a}_r \cdot \bar{a}_R) A_R + (\vec{a}_r \cdot \bar{a}_\theta) A_\theta + (\vec{a}_r \cdot \bar{a}_\phi) A_\phi$
 $= A_R \sin \theta + A_\theta \cos \theta$
 $= \frac{A_R r}{\sqrt{r^2 + z^2}} + \frac{A_\theta z}{\sqrt{r^2 + z^2}}$

P. 2-14 a) In Cartesian coordinates, $\vec{E} = \bar{a}_x E_x + \bar{a}_y E_y + \bar{a}_z E_z$.

$$E_\theta = \bar{a}_\theta \cdot \vec{E} = (\bar{a}_\theta \cdot \bar{a}_x) E_x + (\bar{a}_\theta \cdot \bar{a}_y) E_y + (\bar{a}_\theta \cdot \bar{a}_z) E_z \\ = E_x \cos \theta_1 \cos \phi_1 + E_y \cos \theta_1 \sin \phi_1 - E_z \sin \theta_1.$$

b) In cylindrical coordinates, $\vec{E} = \bar{a}_r E_r + \bar{a}_\phi E_\phi + \bar{a}_z E_z$.

$$E_\theta = \bar{a}_\theta \cdot \vec{E} = (\bar{a}_\theta \cdot \bar{a}_r) E_r + (\bar{a}_\theta \cdot \bar{a}_\phi) E_\phi + (\bar{a}_\theta \cdot \bar{a}_z) E_z \\ = E_r \cos \theta_1 - E_z \sin \theta_1.$$

P. 2-15 a) $\vec{F}_P = \bar{a}_R \frac{12}{\sqrt{(-2)^2 + (-4)^2 + 4^2}} = \bar{a}_R \frac{12}{6} = \bar{a}_R 2.$

$$(\vec{F}_P)_y = 2 \left(\frac{-4}{\sqrt{(-2)^2 + (-4)^2 + 4^2}} \right) = -\frac{4}{3}$$

b) $\bar{a}_F = \frac{1}{6} (-\bar{a}_x 2 - \bar{a}_y 4 + \bar{a}_z 4) = \frac{1}{3} (-\bar{a}_x - \bar{a}_y 2 + \bar{a}_z 2).$

$$\bar{a}_A = \frac{1}{\sqrt{2^2 + (-3)^2 + (-6)^2}} (\bar{a}_x 2 - \bar{a}_y 3 - \bar{a}_z 6) = \frac{1}{7} (\bar{a}_x 2 - \bar{a}_y 3 - \bar{a}_z 6).$$

$$\theta_{FA} = \cos^{-1} (\bar{a}_F \cdot \bar{a}_A) = \cos^{-1} \frac{1}{21} (-2 + 6 \cdot 12) = \cos^{-1} \left(\frac{-8}{21} \right) \\ = \cos^{-1} (-0.381) = 180^\circ - 67.6^\circ = 112.4^\circ.$$

P. 2-16 $\int_P^{P_2} \vec{E} \cdot d\vec{L} = \int_P^{P_2} (y dx + x dy).$

a) $x = 2y^2, dx = 4y dy; \int_P^{P_2} \vec{E} \cdot d\vec{L} = \int_1^2 (4y^2 dy + 2y^2 dy) = 14.$

b) $x = 6y - 4, dx = 6 dy; \int_P^{P_2} \vec{E} \cdot d\vec{L} = \int_1^2 [6y dy + (6y - 4)] dy = 14.$

Equal line integrals along two specific paths do not necessarily imply a conservative field. $\vec{E}$ is a conservative field in this case because $\vec{E} = \nabla(xy + c).$

P. 2-17 a) $\bar{R} = \bar{a}_x x + \bar{a}_y y + \bar{a}_z z, \quad \frac{1}{R} = (x^2 + y^2 + z^2)^{-1/2}$

$$\nabla \left(\frac{1}{R} \right) = \bar{a}_x \frac{\partial}{\partial x} \left(\frac{1}{R} \right) + \bar{a}_y \frac{\partial}{\partial y} \left(\frac{1}{R} \right) + \bar{a}_z \frac{\partial}{\partial z} \left(\frac{1}{R} \right) \\ = -\frac{1}{R^3} (\bar{a}_x x + \bar{a}_y y + \bar{a}_z z) = -\bar{R}/R^3.$$

b) $\bar{R} = \bar{a}_R R, \quad \nabla \left(\frac{1}{R} \right) = \bar{a}_R \frac{\partial}{\partial R} \left(\frac{1}{R} \right) = -\bar{a}_R \left(\frac{1}{R^2} \right) = -\bar{R}/R^3.$

P.2-18 a) $\vec{\nabla} V = \bar{a}_x (2y+z) + \bar{a}_y (2x-z) + \bar{a}_z (x-y)$
 $= \bar{a}_x (-2) + \bar{a}_y 4 + \bar{a}_z 3$; Magnitude $= \sqrt{29}$.

b) $\vec{PQ} = \vec{OQ} - \vec{OP} = \bar{a}_x (-2) + \bar{a}_y 3 + \bar{a}_z 6$,
 $\bar{a}_{PQ} = \frac{\vec{PQ}}{\sqrt{(-2)^2 + 3^2 + 6^2}} = \frac{1}{7} (-\bar{a}_x 2 + \bar{a}_y 3 + \bar{a}_z 6)$.

Rate of increase of V from P toward $Q = (\vec{\nabla} V) \cdot \bar{a}_{PQ}$
 $= \frac{1}{7} (4 + 12 + 18) = \frac{34}{7}$.

P.2-19 a) $\frac{\partial \bar{a}_r}{\partial \phi} = \bar{a}_\phi$; $\frac{\partial \bar{a}_\phi}{\partial \phi} = -\bar{a}_r$.

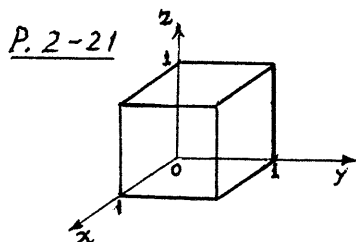
b) $\vec{\nabla} \cdot \bar{A} = (\bar{a}_r \frac{\partial}{\partial r} + \bar{a}_\phi \frac{1}{r} \frac{\partial}{\partial \phi} + \bar{a}_z \frac{\partial}{\partial z}) \cdot (\bar{a}_r A_r + \bar{a}_\phi A_\phi + \bar{a}_z A_z)$
 $= \frac{\partial A_r}{\partial r} + \bar{a}_\phi \frac{1}{r} \cdot \frac{\partial}{\partial \phi} (\bar{a}_r A_r) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$
 $= \frac{\partial A_r}{\partial r} + \bar{a}_\phi \frac{1}{r} \cdot (\bar{a}_r \frac{\partial A_r}{\partial \phi} + A_r \frac{\partial \bar{a}_r}{\partial \phi}) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$
 $= \frac{\partial A_r}{\partial r} + \frac{A_r}{r} + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$
 $= \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$.

P.2-20 In spherical coordinates,

$$\vec{\nabla} \cdot \bar{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R), \text{ if } \bar{A} = \bar{a}_R A_R.$$

a) $\bar{A} = f_1(\bar{R}) = \bar{a}_R R^n$, $A_R = R^n$.
 $\vec{\nabla} \cdot \bar{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^{n+2}) = (n+2) R^{n-1}$.

b) $\bar{A} = f_2(\bar{R}) = \bar{a}_R \frac{k}{R^2}$, $A_R = k R^{-2}$.
 $\vec{\nabla} \cdot \bar{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (k) = 0$.



$\bar{F} = \bar{a}_x xy + \bar{a}_y yz + \bar{a}_z zx$. To find $\oint \bar{F} \cdot d\bar{s}$:

a) Left face: $y=0$, $d\bar{s} = -\bar{a}_y dx dz$.

$$\int_0^1 \int_0^1 -yz dx dz = 0. \quad (1)$$

Right face: $y=1$, $d\vec{s} = \vec{a}_y dx dz$.

$$\int_0^1 \int_0^1 z dx dz = \frac{1}{2}. \quad (2)$$

Top face: $z=1$, $d\vec{s} = \vec{a}_z dx dy$.

$$\int_0^1 \int_0^1 dx dy = \frac{1}{2}. \quad (3)$$

Bottom face: $z=0$, $d\vec{s} = -\vec{a}_z dx dy$, $\int \vec{F} \cdot d\vec{s} = 0$. (4)

Front face: $x=1$, $d\vec{s} = \vec{a}_x dy dz$.

$$\int_0^1 \int_0^1 y dy dz = \frac{1}{2}. \quad (5)$$

Back face: $x=0$, $d\vec{s} = -\vec{a}_x dy dz$, $\int \vec{F} \cdot d\vec{s} = 0$. (6)

Adding the results in (1), (2), (3), (4), (5), and (6):

$$\oint \vec{F} \cdot d\vec{s} = \frac{3}{2}.$$

b) $\vec{\nabla} \cdot \vec{F} = y + z + x$, $dv = dx dy dz$.

$$\int \vec{\nabla} \cdot \vec{F} dv = \int_0^1 \int_0^1 \int_0^1 (x + y + z) dx dy dz = \frac{3}{2}.$$

P. 2-22 $\vec{A} = \vec{a}_r r^2 + \vec{a}_z 2z$.

$$\oint_S \vec{A} \cdot d\vec{s} = \left(\int_{\text{top face}} + \int_{\text{bottom face}} + \int_{\text{walls}} \right) \vec{A} \cdot d\vec{s}.$$

Top face ($z=4$): $\vec{A} = \vec{a}_r r^2 + \vec{a}_z 8$, $d\vec{s} = \vec{a}_z ds$.

$$\int_{\text{top face}} \vec{A} \cdot d\vec{s} = \int_{\text{top face}} 8 ds = 8(\pi 5^2) = 200\pi.$$

Bottom face ($z=0$): $\vec{A} = \vec{a}_r r^2$, $d\vec{s} = -\vec{a}_z ds$, $\int_{\text{bottom face}} \vec{A} \cdot d\vec{s} = 0$.

Walls ($r=5$): $\vec{A} = \vec{a}_r 25 + \vec{a}_z 2z$, $d\vec{s} = \vec{a}_r ds$.

$$\int_{\text{walls}} \vec{A} \cdot d\vec{s} = 25 \int_{\text{walls}} ds = 25(2\pi 5 \times 4) = 1000\pi.$$

$$\therefore \oint \vec{A} \cdot d\vec{s} = 200\pi + 0 + 1000\pi = 1,200\pi.$$

$$\vec{\nabla} \cdot \vec{A} = 3r + 2, \quad \int_V \vec{\nabla} \cdot \vec{A} dv = \int_0^4 \int_0^{2\pi} \int_0^5 (3r + 2) r dr d\phi dz = 1,200\pi.$$

$$= \oint \vec{A} \cdot d\vec{s}.$$

P. 2-23 $\bar{A} = \bar{a}_z z = \bar{a}_z R \cos \theta$.

a) Over the hemispherical surface: $d\bar{s} = \bar{a}_R R^2 \sin \theta d\theta d\phi$.

$$\begin{aligned} \int \bar{A} \cdot d\bar{s} &= \int_0^{\pi/2} \int_0^{2\pi} \bar{a}_z (R \cos \theta) \cdot \bar{a}_R R^2 \sin \theta d\theta d\phi \\ &= R^3 2\pi \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta = \frac{2}{3} \pi R^3. \end{aligned}$$

Over the flat base: $z=0$, $\bar{A}=0$, $\int \bar{A} \cdot d\bar{s} = 0$.

$$\therefore \oint \bar{A} \cdot d\bar{s} = \frac{2}{3} \pi R^3.$$

b) $\bar{\nabla} \cdot \bar{A} = \frac{\partial A_z}{\partial z} = \frac{\partial z}{\partial z} = 1$.

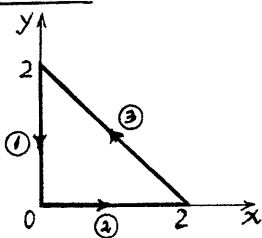
c) $\int \bar{\nabla} \cdot \bar{A} dv = 1 \times (\text{volume of hemispherical region}) = \frac{2}{3} \pi R^3$
 $= \oint \bar{A} \cdot d\bar{s} \rightarrow \text{Divergence theorem is proved.}$

P. 2-24 $\bar{D} = \bar{a}_R \frac{\cos^2 \phi}{R^3}$. $d\bar{s} = \begin{cases} \bar{a}_R R^2 \sin \theta d\theta d\phi, & \text{at } R=3. \\ -\bar{a}_R R^2 \sin \theta d\theta d\phi, & \text{at } R=2. \end{cases}$

a) $\oint \bar{D} \cdot d\bar{s} = \int_0^{2\pi} \int_0^\pi \left(\frac{1}{3} - \frac{1}{2} \right) \sin \theta d\theta \cdot \cos^2 \phi d\phi$
 $= -\frac{1}{6} \int_0^\pi \sin \theta d\theta \int_0^{2\pi} \cos^2 \phi d\phi = -\frac{1}{6} (2) \pi = -\frac{\pi}{3}.$

b) $\bar{\nabla} \cdot \bar{D} = -\frac{\cos^2 \phi}{R^4}$, $dv = R^2 \sin \theta dR d\theta d\phi$.
 $\int \bar{\nabla} \cdot \bar{D} dv = \int_0^{2\pi} \int_0^\pi \int_2^3 \left(-\frac{\cos^2 \phi}{R^2} \right) \sin \theta dR d\theta d\phi = -\frac{\pi}{3}.$

P. 2-26



a) $d\bar{\ell} = \bar{a}_x dx + \bar{a}_y dy$,

$$\bar{A} \cdot d\bar{\ell} = (2x^2 + y^2) dx + (xy - y^2) dy.$$

Path ①: $x=0$, $dx=0$, $\int \bar{A} \cdot d\bar{\ell} = -\int_2^0 y^2 dy = 8/3$.

Path ②: $y=0$, $dy=0$, $\int \bar{A} \cdot d\bar{\ell} = \int_0^2 2x^2 dx = 16/3$.

Path ③: $y=2-x$, $dy=-dx$, $\int \bar{A} \cdot d\bar{\ell} = -28/3$.

$$\oint \bar{A} \cdot d\bar{\ell} = \frac{8}{3} + \frac{16}{3} - \frac{28}{3} = -\frac{4}{3}.$$

b) $\bar{\nabla} \times \bar{A} = -\bar{a}_z y$, $d\bar{s} = \bar{a}_z dx dy$, $\int (\bar{\nabla} \times \bar{A}) \cdot d\bar{s} = -\int_0^2 \left[\int_0^{2-x} y dy \right] dx = -\frac{4}{3}.$

c) No. $\bar{\nabla} \times \bar{A} \neq 0$.

P.2-27 $\vec{F} = \bar{a}_r 5r \sin \phi + \bar{a}_\phi r^2 \cos \phi$.

a) Path AB: $r=1$, $\vec{F} = \bar{a}_r 5 \sin \phi + \bar{a}_\phi \cos \phi$; $d\vec{\ell} = \bar{a}_\phi d\phi$.

$$\int_{AB} \vec{F} \cdot d\vec{\ell} = \int_0^{\pi/2} \cos \phi d\phi = 1.$$

Path BC: $\phi = \pi/2$, $\vec{F} = \bar{a}_r 5r$; $d\vec{\ell} = \bar{a}_r dr$.

$$\int_{BC} \vec{F} \cdot d\vec{\ell} = \int_1^2 5r dr = 15/2.$$

Path CD: $r=2$, $\vec{F} = \bar{a}_r 10 \sin \phi + \bar{a}_\phi 4 \cos \phi$; $d\vec{\ell} = \bar{a}_\phi 2 d\phi$.

$$\int_{CD} \vec{F} \cdot d\vec{\ell} = \int_{\pi/2}^0 8 \cos \phi d\phi = -8.$$

Path DA: $\phi = 0$, $\vec{F} = \bar{a}_\phi r^2$; $d\vec{\ell} = \bar{a}_r dr$.

$$\int_{DA} \vec{F} \cdot d\vec{\ell} = 0.$$

$$\therefore \oint_{ABCD} \vec{F} \cdot d\vec{\ell} = 1 + \frac{15}{2} - 8 = \frac{1}{2}.$$

b) $\vec{\nabla} \times \vec{F} = \bar{a}_z \frac{1}{r} \left[\frac{\partial}{\partial r} (r F_\phi) - \frac{\partial F_r}{\partial \phi} \right] = \bar{a}_z (3r - 5) \cos \phi$.

c) $d\vec{s} = -\bar{a}_z r dr d\phi$, $(\vec{\nabla} \times \vec{F}) \cdot d\vec{s} = -r(3r - 5) dr \cos \phi d\phi$.

$$\int (\vec{\nabla} \times \vec{F}) \cdot d\vec{s} = - \int_1^2 r(3r - 5) dr \int_0^{\pi/2} \cos \phi d\phi = \frac{1}{2}.$$

P.2-28 $\vec{A} = \bar{a}_\phi 3 \sin(\phi/2)$.

$$\vec{\nabla} \times \vec{A} = \frac{3}{2 \sin \theta} (\bar{a}_r \cos \theta \sin \frac{\phi}{2} - \bar{a}_\theta \sin \theta \sin \frac{\phi}{2}).$$

Assume the hemispherical bowl to be located in the lower half of the xy-plane and its circular rim coincident with the xy-plane. Tracing the rim in a counterclockwise direction, we have

$$d\vec{\ell} = \bar{a}_\phi 4 d\phi, d\vec{s} = -\bar{a}_r 4^2 \sin \theta d\theta d\phi.$$

$$\oint_C \vec{A} \cdot d\vec{\ell} = \int_0^{2\pi} (\vec{A})_{\substack{R=4 \\ \theta=\pi/2}} \cdot (\bar{a}_\phi 4 d\phi) = \int_0^{2\pi} 12 \sin(\frac{\phi}{2}) d\phi = 48.$$

$$\int_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s} = -12 \int_0^{2\pi} \int_{\pi/2}^{\pi} \cos \theta \sin \frac{\phi}{2} d\theta d\phi = 48. \\ = \oint_C \vec{A} \cdot d\vec{\ell}.$$

P.2-30. $\vec{F} = \bar{a}_x(x+3y-c_1z) + \bar{a}_y(c_2x+5z) + \bar{a}_z(2x-c_3y+c_4z)$.

a) $\vec{F}$ is irrotational:

$$\vec{\nabla} \times \vec{F} = \bar{a}_x \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \bar{a}_y \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \bar{a}_z \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) = 0.$$

Each component must vanish.

$$\frac{\partial}{\partial y}(2x - c_3y + c_4z) - \frac{\partial}{\partial z}(c_2x + 5z) = 0 \longrightarrow c_3 = -5.$$

$$\frac{\partial}{\partial z}(x + 3y - c_1z) - \frac{\partial}{\partial x}(2x - c_3y + c_4z) = 0 \longrightarrow c_1 = -2.$$

$$\frac{\partial}{\partial x}(c_2x + 5z) - \frac{\partial}{\partial y}(x + 3y - c_1z) = 0 \longrightarrow c_2 = 3.$$

b) F is also solenoidal:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 0.$$

$$\frac{\partial}{\partial x}(x + 3y - c_1z) + \frac{\partial}{\partial y}(c_2x + 5z) + \frac{\partial}{\partial z}(2x - c_3y + c_4z) = 0.$$

$$\longrightarrow c_4 = -1.$$

Chapter 3

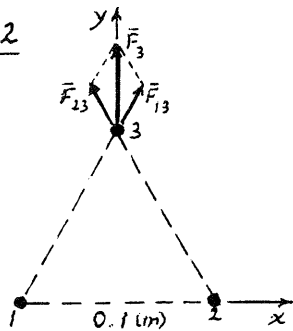
Static Electric Fields

- P. 3-1 a) Max. voltage $V_{\max}$ will make $d_1 = h/2$ at $z = w$

$$\frac{h}{2} = \frac{e}{2m} \left(\frac{V_{\max}}{h} \right) \left(\frac{w}{u_0} \right)^2 \rightarrow V_{\max} = \frac{m}{e} \left(\frac{u_0 h}{w} \right)^2$$
- b) At the screen, $(d_0)_{\max} = D/2$. L must be $\leq L_{\max}$, where

$$L_{\max} = \frac{1}{2} \left(w + \frac{m u_0^2 D h}{e w V_{\max}} \right)$$
- c) Double $V_{\max}$ by doubling u_0^2 , or doubling the anode accelerating voltage.

P. 3-2



$$\vec{F}_{13} = \frac{(2 \times 10^{-6})^2}{4\pi\epsilon_0 (0.1)^2} (\bar{a}_x 0.5 + \bar{a}_y 0.866)$$

$$= 3.6 (\bar{a}_x 0.5 + \bar{a}_y 0.866) \text{ (N)}$$

$$\vec{F}_{23} = 3.6 (-\bar{a}_x 0.5 + \bar{a}_y 0.866) \text{ (N)}$$

$$\vec{F}_3 = \vec{F}_{13} + \vec{F}_{23} = \bar{a}_y 0.624 \text{ (N)}$$

!6.24N olacak

Similarly for $\vec{F}_1$ and $\vec{F}_2$. All are repulsive forces in the direction away from the center of the triangle.

P. 3-3 $\vec{Q}_1 \vec{P} = -\bar{a}_y 3 + \bar{a}_z 4$, $\vec{Q}_2 \vec{P} = \bar{a}_y 4 - \bar{a}_z 3$.

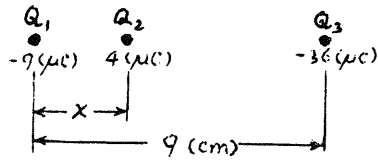
At P: $\vec{E}_1 = \frac{Q_1}{4\pi\epsilon_0 (5)^3} (-\bar{a}_y 3 + \bar{a}_z 4)$,

$$\vec{E}_2 = \frac{Q_2}{4\pi\epsilon_0 (5)^3} (\bar{a}_y 4 - \bar{a}_z 3)$$

a) No y-component: $-3Q_1 + 4Q_2 = 0 \rightarrow \frac{Q_1}{Q_2} = \frac{4}{3}$

b) No z-component: $4Q_1 - 3Q_2 = 0 \rightarrow \frac{Q_1}{Q_2} = \frac{3}{4}$

P. 3-4



For zero force on Q_1 :

$$\frac{Q_1 Q_2}{4\pi\epsilon_0 x^2} + \frac{Q_1 Q_3}{4\pi\epsilon_0 9^2} = 0.$$

$$x = 9\sqrt{\frac{Q_2}{Q_3}} = 9\sqrt{\frac{4}{36}} = 3 \text{ (cm)}.$$

With $x = 3 \text{ (cm)}$, it can be proved that the net forces on Q_2 and Q_3 are also zero.

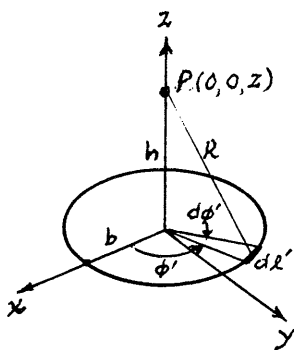
P. 3-5 From Eq. (3-42a), $\rho_s = \frac{\text{Total charge}}{\text{Disk area}} = \frac{Q}{\pi b^2}.$

$$\begin{aligned}\bar{E} &= \bar{a}_z \frac{\rho_s}{2\epsilon_0} \left[1 - \left(1 + \frac{b^2}{z^2} \right)^{-1/2} \right] \\ &= \bar{a}_z \frac{\rho_s}{2\epsilon_0} \left[1 - \left(1 - \frac{b^2}{2z^2} + \frac{3}{8} \frac{b^4}{z^4} - \dots \right) \right] \\ &= \bar{a}_z \frac{Q}{2\pi\epsilon_0 b^2} \left[\frac{1}{2} \left(\frac{b}{z} \right)^2 - \frac{3}{8} \left(\frac{b}{z} \right)^4 + \dots \right] \\ &= \bar{a}_z \left[\frac{Q}{4\pi\epsilon_0 z^2} \left(1 - \frac{3}{4} \frac{b^2}{z^2} + \dots \right) \right],\end{aligned}$$

where the first term is the point-charge term and the rest represent the error. Considering only the first error term:

$$\frac{3}{4} \cdot \frac{b^2}{z^2} \leq 0.01 \rightarrow z \geq \sqrt{75} b, \text{ or } 8.66 b.$$

P. 3-6 At an arbitrary $P(0,0,z)$ on the axis:



$$dV_p = \frac{\rho_l b d\phi'}{4\pi\epsilon_0 (z^2 + b^2)^{1/2}}.$$

$$V_p = \frac{\rho_l b}{4\pi\epsilon_0 (z^2 + b^2)^{1/2}} \int_0^{2\pi} d\phi' = \frac{\rho_l b}{2\epsilon_0 (z^2 + b^2)^{1/2}}.$$

$$\bar{E}_p = -\bar{\nabla} V_p = -\bar{a}_z \frac{dV_p}{dz} = \bar{a}_z \frac{\rho_l b}{2\epsilon_0 (z^2 + b^2)^{3/2}}. \quad \text{!çarpı } z \text{ olacak}$$

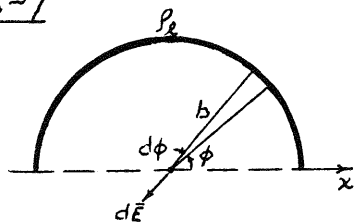
$$a) \text{ At point } (0,0,h), \bar{E} = \bar{a}_z \frac{\rho_l b}{2\epsilon_0 (h^2 + b^2)^{3/2}}. \quad \text{!çarpı } h \text{ olacak}$$

$$b) \text{ To find the location of max. } |\bar{E}_p|, \text{ set } \frac{\partial}{\partial z} |\bar{E}_p| = 0 \rightarrow z = \frac{b}{\sqrt{2}}. \text{ Max } |\bar{E}_p| = \frac{\rho_l}{3.67\epsilon_0 b^2}.$$

Similar situation when P is below the loop.

!paydası
5.196epsilon sıfır
çarpı b olacak

P.3-7



$$dE_y = -\frac{\rho_l(b d\phi)}{4\pi\epsilon_0 b^2} \sin\phi,$$

$$\vec{E} = \vec{a}_y E_y = -\vec{a}_y \frac{\rho_l}{4\pi\epsilon_0 b} \int_0^\pi \sin\phi d\phi$$

$$= -\vec{a}_y \frac{\rho_l}{2\pi\epsilon_0 b}.$$

P.3-8 Spherical symmetry: $\vec{E} = \vec{a}_R E_R$. Apply Gauss's law.

$$1) 0 \leq R \leq b. \quad 4\pi R^2 E_{R1} = \frac{\rho_0}{\epsilon_0} \int_0^R \left(1 - \frac{R'^2}{b^2}\right) 4\pi R'^2 dR' = \frac{4\pi\rho_0}{\epsilon_0} \left(\frac{R^3}{3} - \frac{R^5}{5b^2}\right),$$

$$E_{R1} = \frac{\rho_0 R}{\epsilon_0} \left(\frac{1}{3} - \frac{R^2}{5b^2}\right).$$

$$2) b \geq R < R_i. \quad 4\pi R^2 E_{R2} = \frac{\rho_0}{\epsilon_0} \int_0^b \left(1 - \frac{R'^2}{b^2}\right) 4\pi R'^2 dR' = \frac{8\pi\rho_0}{15\epsilon_0} b^3,$$

$$E_{R2} = \frac{2\rho_0 b^3}{15\epsilon_0 R^2}.$$

$$3) R_i < R < R_o. \quad E_{R3} = 0.$$

$$4) R > R_o. \quad E_{R4} = \frac{2\rho_0 b^3}{15\epsilon_0 R^2}.$$

P.3-9 Cylindrical symmetry: $\vec{E} = \vec{a}_r E_r$. Apply Gauss's law.

$$a) E_r = 0, \text{ for } r < a.$$

$$E_r = \frac{a\rho_{sa}}{\epsilon_0 r}, \text{ for } a < r < b.$$

$$E_r = \frac{a\rho_{sa} + b\rho_{sb}}{\epsilon_0 r}, \text{ for } r > b.$$

$$b) \frac{b}{a} = -\frac{\rho_{sa}}{\rho_{sb}}.$$

P.3-10 $W_e = -q \int \vec{E} \cdot d\vec{l} = -q \int (y dx + x dy).$

a) Along the parabola $y = 2x^2$, $dy = 4x dx$.

$$W_e = -(5 \times 10^{-6}) \int_1^{-2} (2x^2 + 4x^2) dx = 9 \times 10^{-5} (J) = 90 (\mu J).$$

b) Along the straight line $\frac{y-2}{x-1} = \frac{8-2}{-2-1} = -2$, $y = -2x + 4$, $dy = -2dx$.

$$W_e = -(5 \times 10^{-6}) \int_1^{-2} [(-2x + 4) dx - 2x dx] = 90 (\mu J).$$

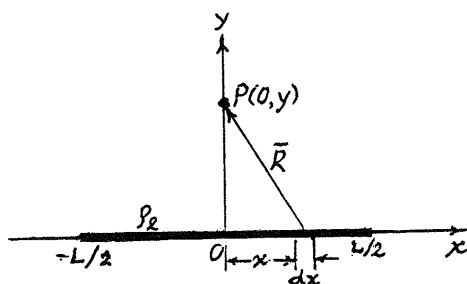
P.3-11 $\vec{E} = \bar{a}_x y - \bar{a}_y x$, $\vec{E} \cdot d\vec{l} = y dx - x dy$.

a) $W_e = -q \int_1^{-2} (2x^2 - 4x^2) dx = -30 (\mu J)$.

b) $W_e = -q \int_1^{-2} [(-2x+4)+2x] dx = -60 (\mu J)$.

The given $\vec{E}$ field is nonconservative.

P.3-12



a)
$$V = 2 \int_0^{L/2} \frac{\rho_l dx}{4\pi\epsilon_0 R}$$

$$= \frac{\rho_l}{2\pi\epsilon_0} \int_0^{L/2} \frac{dx}{\sqrt{x^2 + y^2}}$$

$$= \frac{\rho_l}{2\pi\epsilon_0} \left\{ \ln \left[\sqrt{\left(\frac{L}{2}\right)^2 + y^2} + \frac{L}{2} \right] - \ln y \right\}.$$

b) From Coulomb's law:

$$\vec{E} = \bar{a}_y E_y = 2 \int_0^{L/2} \bar{a}_y \frac{\rho_l y dx}{4\pi\epsilon_0 R^3} = \bar{a}_y \frac{\rho_l}{2\pi\epsilon_0 y} \left[\frac{L/2}{\sqrt{(L/2)^2 + y^2}} \right].$$

c) $\vec{E} = -\nabla V$ gives the same answer as in b).

P.3-13 a) $\rho_{ps} = \vec{P} \cdot \bar{a}_n = P_0 \frac{L}{2}$ on all six faces of the cube.

$$\rho_{pv} = -\vec{\nabla} \cdot \vec{P} = -3P_0.$$

b) $Q_s = (6L^2) \rho_{ps} = 3P_0 L^3$, $Q_v = (L^3) \rho_{pv} = -3P_0 L^3$.

$$\text{Total bound charge} = Q_s + Q_v = 0.$$

P.3-14 $\vec{P} = \bar{a}_x P_0$.

a) $\rho_{ps} = \vec{P} \cdot \bar{a}_n = P_0 \sin \theta \cos \phi$.

$$\rho_{pv} = -\vec{\nabla} \cdot \vec{P} = 0.$$

b) $Q_s = \int_0^\pi \int_0^{2\pi} P_0 b^2 \sin^2 \theta \cos \phi d\phi d\theta$
 $= 0.$

P.3-15 $\bar{P} = P_0 (\bar{a}_x 3x + \bar{a}_y 4y)$.

a) $\rho_{pr} = -\bar{\nabla} \cdot \bar{P} = -7P_0$.

Total volume charge $Q_v = -7P_0 \pi (r_o^2 - r_i^2)$ per unit length.

Outer surface: $r = r_o, \bar{a}_n = \bar{a}_r, \rho_{ps_o} = \bar{P} \cdot \bar{a}_r = P_0 (\bar{a}_x 3r_o \cos \phi + \bar{a}_y 4r_o \sin \phi) \cdot \bar{a}_r$
 $= P_0 r_o (3 \cos^2 \phi + 4 \sin^2 \phi)$
 $= P_0 r_o (3 + \sin^2 \phi)$.

Inner surface: $r = r_i, \bar{a}_n = -\bar{a}_r, \rho_{ps_i} = -P_0 r_i (3 + \sin^2 \phi)$.

b) Total $Q_{s_o} = \int_0^{2\pi} \rho_{ps_o} r_o d\phi = P_0 r_o^2 \int_0^{2\pi} (3 + \sin^2 \phi) d\phi = 7\pi P_0 r_o^2$,
 per unit length.

Total $Q_{s_i} = -7\pi P_0 r_i^2$, per unit length.

Total bound charge: $Q_v + Q_{s_o} + Q_{s_i} = 0$.

P.3-16 Spherical symmetry: Apply Gauss's law. $\bar{E} = \bar{a}_R E_R, \bar{D} = \bar{a}_R D_R$.

(1) $R > R_o, E_{R1} = \frac{Q}{4\pi\epsilon_0 R^2}, V_1 = \frac{Q}{4\pi\epsilon_0 R}$,

$D_{R1} = \epsilon_0 E_{R1} = \frac{Q}{4\pi R^2}, P_{R1} = 0$.

(2) $R_i < R < R_o, E_{R2} = \frac{Q}{4\pi\epsilon_0 \epsilon_r R^2}, D_{R2} = \frac{Q}{4\pi R^2}$,

$P_{R2} = D_{R2} - \epsilon_0 E_{R2} = \left(1 - \frac{1}{\epsilon_r}\right) \frac{Q}{4\pi R^2}$.

! işaret
+ olmalı

$V_2 = -\int_{\infty}^{R_o} E_{R1} dR - \int_{R_o}^R E_{R2} dR = \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \frac{1}{\epsilon_r R} \right]$.

(3) $R < R_i, E_{R3} = \frac{Q}{4\pi\epsilon_0 R^2}, D_{R3} = \frac{Q}{4\pi R^2}, P_{R3} = 0$.

$V_3 = V_2 \Big|_{R=R_i} - \int_{R_i}^R E_{R3} dR$
 $= \frac{Q}{4\pi\epsilon_0} \left[\left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_o} - \left(1 - \frac{1}{\epsilon_r}\right) \frac{1}{R_i} + \frac{1}{R} \right]$.

P.3-17 Use subscript a for air, p for plexiglass, and b for breakdown.

$$a) V_b = E_{ba} d_a = (3 \times 10^6) \times (50 \times 10^{-3}) = 150 \times 10^3 (V) = 150 (kV).$$

$$b) V_b = E_{bp} d_p = 20 \times 50 = 1,000 (kV).$$

$$c) V_b = E_a d_a + E_p d_p = E_a (50 - d_p) + E_p d_p$$

$$\text{Now } D_a = D_p \rightarrow \epsilon_0 E_a = \epsilon_0 \epsilon_{rp} E_p \rightarrow E_a = \epsilon_{rp} E_p > E_p.$$

$$E_{ba} < E_{bp} \rightarrow \text{Breakdown occurs in air region first.}$$

$$\therefore V_b = E_{ba} (50 - 10) + \frac{E_{ba}}{3} \times 10 = 3(40 - \frac{1}{3} \times 10) = 130 (kV).$$

P.3-18 At the $z=0$ plane: $\vec{E}_1 = \vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z 5$.

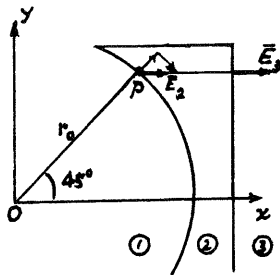
$$\vec{E}_{1t}(z=0) = \vec{E}_{2t}(z=0) = \vec{a}_x 2y - \vec{a}_y 3x.$$

$$\begin{aligned} \vec{D}_{1n}(z=0) &= \vec{D}_{2n}(z=0) \rightarrow 2\vec{E}_{1n}(z=0) = 3\vec{E}_{2n}(z=0), \\ &\rightarrow E_{2n}(z=0) = \frac{2}{3}(\vec{a}_z 5) = \vec{a}_z \frac{10}{3}. \end{aligned}$$

$$\therefore \vec{E}_2(z=0) = \vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z \frac{10}{3}.$$

$$\vec{D}_2(z=0) = (\vec{a}_x 2y - \vec{a}_y 3x + \vec{a}_z \frac{10}{3}) 3\epsilon_0$$

P.3-19



$$\text{Assume } \vec{E}_2 = \vec{a}_r E_{2r} + \vec{a}_\phi E_{2\phi}.$$

$$\text{Boundary condition: } \vec{a}_n \times \vec{E}_1 = \vec{a}_n \times \vec{E}_2.$$

$$\rightarrow E_{2\phi} = -3.$$

For $\vec{E}_3$, and hence $\vec{E}_2$, to be parallel to the x -axis, $E_{2\phi} = -E_{2r}$.

$$\rightarrow E_{2r} = 3.$$

$$\text{Boundary condition: } \vec{a}_n \cdot \vec{D}_1 = \vec{a}_n \cdot \vec{D}_2.$$

$$\rightarrow \epsilon_1 E_{r1} = \epsilon_2 E_{r2} \rightarrow \epsilon_0 5 = \epsilon_0 \epsilon_{r2} 3.$$

$$\rightarrow \epsilon_{r2} = \frac{5}{3} = 1.667.$$

P.3-20 $\epsilon = \frac{\epsilon_2 - \epsilon_1}{d} y + \epsilon_1$.

Assume Q on plate at $y=d$. $\vec{E} = -\vec{a}_y \frac{\rho_s}{\epsilon} = \frac{Q}{s(\frac{\epsilon_2 - \epsilon_1}{d} y + \epsilon_1)}$.

$$V = -\int_{y=0}^{y=d} \vec{E} \cdot d\vec{l} = \frac{Qd \ln(\epsilon_2/\epsilon_1)}{s(\epsilon_2 - \epsilon_1)}.$$

$$C = \frac{Q}{V} = \frac{s(\epsilon_2 - \epsilon_1)}{d \ln(\epsilon_2/\epsilon_1)}.$$

P.3-21 Let ρ_ℓ be the lineal charge density on the inner conductor. $\vec{E} = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon r}$.

$$V_0 = -\int_b^a \vec{E} \cdot d\vec{r} = \frac{\rho_\ell}{2\pi\epsilon} \ln\left(\frac{b}{a}\right) \rightarrow \rho_\ell = \frac{2\pi\epsilon V_0}{\ln(b/a)}.$$

a) $\vec{E}(a) = \vec{a}_r \frac{V_0}{a \ln(b/a)}$.

b) For a fixed b , the function to be minimized is: $(x = b/a)$.

$$f(x) = \frac{V_0 x}{b \ln x}. \text{ Setting } \frac{df(x)}{dx} = 0 \text{ yields } \ln x = 1,$$

$$\text{or } x = \frac{b}{a} = e = 2.718.$$

c) $\min. E(a) = e V_0 / b$.

d) $C' = \frac{\rho_\ell}{V_0} = \frac{2\pi\epsilon}{\ln(b/a)} = 2\pi\epsilon \text{ (F/m)}.$

P.3-22 $\vec{D} = \vec{a}_r \frac{\rho_\ell}{2\pi r}$. $\vec{E}_1 = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r1}r}$, $r_i < r < b$;

$$\vec{E}_2 = \vec{a}_r \frac{\rho_\ell}{2\pi\epsilon_0\epsilon_{r2}r}, \quad b < r < r_o.$$

$$V = -\int_{r_o}^{r_i} \vec{E} \cdot d\vec{r} = \frac{\rho_\ell}{2\pi\epsilon_0} \left[\frac{1}{\epsilon_{r1}} \ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{r_o}{b}\right) \right],$$

$$C' = \frac{\rho_\ell}{V} = \frac{2\pi\epsilon_0}{\frac{1}{\epsilon_{r1}} \ln\left(\frac{b}{r_i}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{r_o}{b}\right)} \text{ (F/m)}.$$

P.3-23 Assume charges $+Q$ and $-Q$ on the inner and outer conductors, respectively. $\vec{E} = \vec{a}_R E_R = \vec{a}_R \frac{Q}{4\pi\epsilon_0 R^2}$.

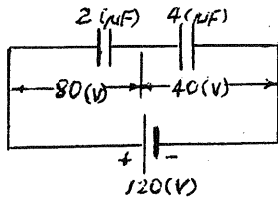
$$V = -\int_{R_o}^{R_i} \vec{E} \cdot \vec{a}_R dR = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_i} - \frac{1}{R_o} \right).$$

$$C = \frac{Q}{V} = \frac{4\pi\epsilon}{(1/R_i - 1/R_o)}.$$

P.3-24 Total capacitance across battery terminals,

$$C_T = \frac{4}{3} (\mu F).$$

Total stored electric energy $W_T = \frac{1}{2} C_T (120)^2 = 9.6 (mJ).$



$$W_e \text{ in } 2(\mu F) \text{ capacitor} = \frac{1}{2} (2 \times 10^{-6}) \times 80^2 = 6.4 (mJ).$$

$$W_e \text{ in } 4(\mu F) \text{ capacitor} = \frac{1}{2} (4 \times 10^{-6}) \times 40^2 = 3.2 (mJ).$$

$$W_e \text{ in } 3(\mu F) \text{ capacitor} = \frac{1}{2} (3 \times 10^{-6}) \times 40^2 = 2.4 (mJ).$$

!Buradaki
birimler
W değil
J=Joule
olmalı

P.3-25 $\vec{E} = \bar{a}_r 6r \sin \phi + \bar{a}_\phi 3r \cos \phi,$

$$d\vec{l} = \bar{a}_r dr + \bar{a}_\phi r d\phi + \bar{a}_z dz,$$

$$W_e = -Q \int_P^B \vec{E} \cdot d\vec{l} = -(5 \times 10^{-10}) \left[6 \sin \phi \int_2^4 r dr + 3r^2 \int_{\pi/3}^{-\pi/2} \cos \phi d\phi \right].$$

a) First $r=2$, ϕ from $\pi/3$ to $-\pi/2$; then $\phi=-\pi/2$, r from 2 to 4:

$$W_e = -(5 \times 10^{-10}) \left[3(2)^2 \sin \phi \Big|_{\pi/3}^{-\pi/2} + 6 \sin(-\frac{\pi}{2}) \frac{r^2}{2} \Big|_2^4 \right]$$

$$= -(5 \times 10^{-10}) [-18 - 36] = 27 \times 10^{-9} (J) = 27 (nJ).$$

!29.2nJ

b) First $\phi=\pi/3$, r from 2 to 4; then $r=4$, ϕ from $\pi/3$ to $-\pi/2$:

$$W_e = -(5 \times 10^{-10}) \left[6 \sin(\frac{\pi}{3}) \frac{r^2}{2} \Big|_2^4 + 3(4)^2 \sin \phi \Big|_{\pi/3}^{-\pi/2} \right]$$

$$= -(5 \times 10^{-10}) [18 - 72] = 27 \times 10^{-9} (J) = 27 (nJ).$$

!29.2nJ

— Same as W_e in part a). $\nabla \times \vec{E} = 0 \rightarrow \vec{E}$ is conservative.

P.3-26 Assume the inner and outer radii to be a and $a+r$ respectively. Substituting Eq.(3-89) in Eq.(3-117) and using Eq.(3-115), we have

$$F_a = -\frac{\partial}{\partial r} \left(\frac{Q}{2} \cdot \frac{Q}{2\pi\epsilon L} \ln \frac{a+r}{a} \right)$$

$$= -\frac{Q^2}{4\pi\epsilon L(a+r)} = -\frac{Q^2}{4\pi\epsilon Lb}, \text{ in the direction of decreasing } r \text{ (attraction).}$$

P.3-27 Switch open: Charges on the plates are constant.

$$Q = CV_0, \quad W_e = \frac{Q^2}{2C}.$$

$$C = \frac{w}{d} [\epsilon x + \epsilon_0 (L-x)].$$

$$\begin{aligned} \bar{F}_e &= -\bar{\nabla} W_e = -\bar{a}_x \frac{Q^2}{2} \frac{\partial}{\partial x} \left(\frac{1}{C} \right) \\ &= \bar{a}_x \frac{Q^2 d}{2w} \frac{\epsilon - \epsilon_0}{[\epsilon x + \epsilon_0 (L-x)]^2} = \bar{a}_x \frac{V_0^2 w}{2d} (\epsilon - \epsilon_0). \end{aligned}$$

P.3-28 Use subscripts d and a to denote dielectric and air regions respectively. $\bar{\nabla}^2 V = 0$ in both regions.

$$V_d = c_1 y + c_2, \quad \bar{E}_d = -\bar{a}_y c_1, \quad \bar{D}_d = -\bar{a}_y \epsilon_0 \epsilon_r c_1.$$

$$V_a = c_3 y + c_4, \quad \bar{E}_a = -\bar{a}_y c_3, \quad \bar{D}_a = -\bar{a}_y \epsilon_0 c_3.$$

$$\text{B.C.: At } y=0, V_d = 0; \quad \text{at } y=d, V_a = V_0;$$

$$\text{at } y=0.8d: V_d = V_a, \quad \bar{D}_d = \bar{D}_a.$$

$$\text{Solving: } c_1 = \frac{V_0}{(0.8+0.2\epsilon_r)d}, \quad c_2 = 0, \quad c_3 = \frac{\epsilon_r V_0}{(0.8+0.2\epsilon_r)d}, \quad c_4 = \frac{(1-\epsilon_r)V_0}{1+0.25\epsilon_r}.$$

$$\text{a) } V_d = \frac{5yV_0}{(4+\epsilon_r)d}, \quad \bar{E}_d = -\bar{a}_y \frac{5V_0}{(4+\epsilon_r)d}.$$

$$\text{b) } V_a = \frac{5\epsilon_r y - 4(\epsilon_r - 1)d}{(4+\epsilon_r)d} V_0, \quad \bar{E}_a = -\bar{a}_y \frac{5\epsilon_r V_0}{(4+\epsilon_r)d}.$$

$$\begin{aligned} \text{c) } (\rho_s)_{y=d} &= -(D_a)_{y=d} = \frac{5\epsilon_0 \epsilon_r V_0}{(4+\epsilon_r)d} \\ (\rho_s)_{y=0} &= (D_d)_{y=0} = -\frac{5\epsilon_0 \epsilon_r V_0}{(4+\epsilon_r)d}. \end{aligned}$$

P.3-29 Poisson's eq. $\bar{\nabla}^2 V = -\frac{A}{\epsilon r} \rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = -\frac{A}{\epsilon r}.$

$$\text{Solution: } V = -\frac{A}{\epsilon} r + c_1 \ln r + c_2.$$

$$\text{B.C.: } \begin{cases} \text{At } r=a, V_0 = -\frac{A}{\epsilon} a + c_1 \ln a + c_2. \\ \text{At } r=b, 0 = -\frac{A}{\epsilon} b + c_1 \ln b + c_2. \end{cases}$$

$$c_1 = \frac{\frac{A}{\epsilon}(b-a) - V_0}{\ln(b/a)},$$

$$c_2 = \frac{V_0 \ln b + \frac{A}{\epsilon}(a \ln b - b \ln a)}{\ln(b/a)}.$$

P.3-30 $\bar{\nabla}^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0.$

Solution: $V = c_1 \ln r + c_2.$

Boundary conditions:

At $r=a$, $V=V_0 = c_1 \ln a + c_2.$

At $r=b$, $V=0 = c_1 \ln b + c_2.$

$c_1 = -\frac{V_0}{\ln(b/a)}, \quad c_2 = \frac{V_0 \ln b}{\ln(b/a)}.$

$V = V_0 \frac{\ln(b/r)}{\ln(b/a)}, \quad \bar{E} = -\bar{\nabla} V = \bar{a}_r \frac{V_0}{r \ln(b/a)}.$

Surface densities: At $r=a$, $\rho_{sa} = \epsilon_0 E_r = \frac{\epsilon_0 V_0}{a \ln(b/a)}.$

At $r=b$, $\rho_{sb} = -\epsilon_0 E_r = -\frac{\epsilon_0 V_0}{b \ln(b/a)}.$

Capacitance per unit length $C' = \frac{Q}{V_{ab}} = \frac{2\pi a \rho_{sa}}{V_0} = \frac{2\pi \epsilon_0}{\ln(b/a)} \quad (\text{C/m}).$

P.3-31 V and $\bar{E}$ depend only on θ . $\rightarrow$ Eq. (3-129): $\frac{d}{d\theta} \left(\sin \theta \frac{dV}{d\theta} \right) = 0.$

a) Solution: $\frac{dV}{d\theta} = \frac{C_1}{\sin \theta} \rightarrow V(\theta) = C_1 \ln \left(\tan \frac{\theta}{2} \right) + C_2.$

B.C. ① $V(\alpha) = V_0 = C_1 \ln \left(\tan \frac{\alpha}{2} \right) + C_2.$

② $V(\frac{\pi}{2}) = 0 = C_1 \ln \left(\tan \frac{\pi}{4} \right) + C_2 \rightarrow C_2 = 0.$

$\rightarrow C_1 = \frac{V_0}{\ln [\tan(\alpha/2)]} \rightarrow V(\theta) = \frac{V_0 \ln [\tan(\theta/2)]}{\ln [\tan(\alpha/2)]}.$

b) $\bar{E} = -\bar{a}_\theta \frac{dV}{R d\theta} = -\bar{a}_\theta \frac{V_0}{R \ln [\tan(\alpha/2)] \sin \theta}.$

c) On the cone $\theta = \alpha$, $\rho_s = \epsilon_0 E(\alpha) = -\frac{\epsilon_0 V_0}{R \ln [\tan(\alpha/2)] \sin \alpha}.$

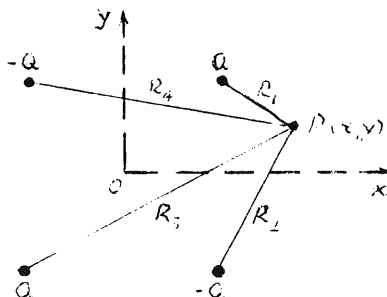
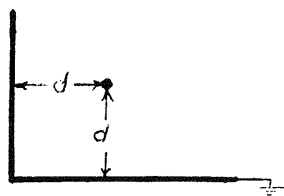
On the grounded plane; $\theta = \pi/2$, $\rho_s = -\epsilon_0 E(\frac{\pi}{2}) = \frac{\epsilon_0 V_0}{R \ln [\tan(\alpha/2)]}.$

P.3-32 Consider the conditions in the xy -plane ($z=0$).

a) $V_p = \frac{Q}{4\pi\epsilon} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3} - \frac{1}{R_4} \right)$, where

$R_1 = [(x-d)^2 + (y-d)^2]^{1/2}, \quad R_2 = [(x-d)^2 + (y+d)^2]^{1/2},$

$R_3 = [(x+d)^2 + (y+d)^2]^{1/2}, \quad R_4 = [(x+d)^2 + (y-d)^2]^{1/2}.$



$$\begin{aligned}\vec{E}_P &= -\vec{\nabla} V_P = -\vec{a}_x \frac{\partial V_P}{\partial x} - \vec{a}_y \frac{\partial V_P}{\partial y} \\ &= \vec{a}_x \frac{Q}{4\pi\epsilon} \left[-\frac{x-d}{R_1^3} + \frac{x-d}{R_2^3} - \frac{x+d}{R_3^3} + \frac{x+d}{R_4^3} \right] \\ &\quad + \vec{a}_y \frac{Q}{4\pi\epsilon} \left[-\frac{y-d}{R_1^3} + \frac{y+d}{R_2^3} - \frac{y+d}{R_3^3} + \frac{y-d}{R_4^3} \right].\end{aligned}$$

E_P will have a z -component if the point P does not lie in the xy -plane.

b) On the conducting half-planes, $\rho_s = D_n = \epsilon E_n$.

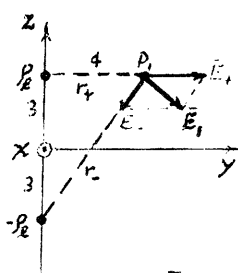
Along the x -axis, $y=0$; $R_1 = [(x-d)^2 + d^2]^{1/2} = R_2$,
and $R_3 = [(x+d)^2 + d^2]^{1/2} = R_4$.

$$E_x = 0, \quad E_y = \frac{Q}{2\pi\epsilon} \left[\frac{d}{R_1^3} - \frac{d}{R_3^3} \right].$$

$$\begin{aligned}\therefore \rho_s(y=0) &= \frac{Qd}{2\pi} \left\{ \frac{1}{[(x-d)^2 + d^2]^{3/2}} - \frac{1}{[(x+d)^2 + d^2]^{3/2}} \right\} \\ &= \begin{cases} 0, & \text{at } x=0. \\ \text{max.}, & \text{at } x=d. \end{cases}\end{aligned}$$

Similarly for $\rho_s(x=0)$ on the vertical conducting plane by changing x to y .

P.3-34



Assume ρ_l (50 nC/m) to be at $y=0$ and $z=3$ (m)

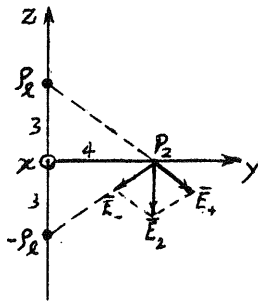
a) Vector from ρ_l to $P_1(0, 4, 3)$: $\vec{r}_+ = \vec{a}_y 4$.

Vector from $-\rho_l$ to P_1 : $\vec{r}_- = \vec{a}_y 4 + \vec{a}_z 6$.

$$\vec{E}_+ = \frac{\rho_l}{2\pi\epsilon_0} \frac{\vec{r}_+}{r_+^2} = \frac{\rho_l}{2\pi\epsilon_0} \frac{\vec{a}_y}{4} = 9 \times 10^5 (\vec{a}_y 0.25)$$

$$\vec{E}_- = \frac{-\rho_l}{2\pi\epsilon_0} \frac{\vec{r}_-}{r_-^2} = -9 \times 10^5 (\vec{a}_y 0.077 + \vec{a}_z 0.115)$$

$$\vec{E}_1 = \vec{E}_+ + \vec{E}_- = 9 \times 10^5 (\vec{a}_y 0.173 - \vec{a}_z 0.115) \text{ (V/m) at } P_1.$$



b) At $P_2 (0, 4, 0)$ on the xy -plane (the ground):

Vector from P_1 to P_2 is $\vec{r}_+ = \vec{a}_y 4 - \vec{a}_z 3$,

Vector from $-P_1$ to P_2 is $\vec{r}_- = \vec{a}_y 4 + \vec{a}_z 3$.

$$\vec{E}_+ = \frac{\rho_L}{2\pi\epsilon_0} \frac{\vec{a}_y 4 - \vec{a}_z 3}{4^2 + 3^2}, \quad \vec{E}_- = \frac{-\rho_L}{2\pi\epsilon_0} \frac{\vec{a}_y 4 + \vec{a}_z 3}{4^2 + 3^2}$$

$$\vec{E}_2 = \vec{E}_+ + \vec{E}_- = \frac{\rho_L}{2\pi\epsilon_0} \left(\frac{-\vec{a}_z 6}{4^2 + 3^2} \right)$$

$$= 9 \times 10^5 (-\vec{a}_z 0.24) = -\vec{a}_z 2.16 \times 10^5 \text{ (V/m)}$$

$$\rho_{s2} = \epsilon_0 E_{2z} = \frac{\rho_L}{2\pi} (-0.24) = \frac{50 \times 10^{-6}}{2\pi} (0.24) = -1.91 \times 10^{-6} \text{ (C/m}^2\text{)} \\ = -1.91 \text{ (}\mu\text{C/m}^2\text{)}.$$

P.3-35 Given $D = 2 \text{ (cm)}$, $a = 0.3 \text{ (cm)}$.

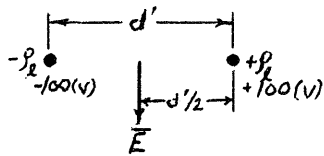
a) From Eq. (3-163),

$$d = \frac{1}{2} (D + \sqrt{D^2 - 4a^2}) = \frac{1}{2} [2 + \sqrt{2^2 - 4(0.3)^2}] = 1.954 \text{ (cm)}$$

$$d_i = D - d = 2 - 1.954 = 0.046 \text{ (cm)} = 0.46 \text{ (mm)}.$$

$$b) \rho_L = \frac{2\pi\epsilon_0 V_i}{\ln(d/a)} = \frac{2\pi(\frac{1}{36\pi} \times 10^{-9}) \times 100}{\ln(1.954/0.3)} = 2.96 \times 10^{-9} \text{ (F/m)} \\ = 2.96 \text{ (nF/m)}.$$

c) The equivalent line charges are separated by



$$d' = d - d_i = 1.954 - 0.046 \\ = 1.908 \text{ (cm)}.$$

$$|\vec{E}| = \frac{\rho_L}{2\pi\epsilon_0 (d'/2)} \times 2 = 111.9 \text{ (V/m)},$$

— in a direction normal to the plane containing the wires.

Chapter 4

Steady Electric Currents

P. 4-1 a) $R = \frac{\ell}{\sigma S} = \frac{V}{I} \rightarrow \sigma = \frac{\ell I}{SV} = 3.54 \times 10^7 \text{ (S/m)}.$

b) $E = \frac{V}{\ell} = 6 \times 10^{-3} \text{ (V/m)}.$

c) $P = VI = 1 \text{ (W)}.$

d) $\rho_e = -\frac{\sigma}{\mu_e}$. The given electron mobility $1.4 \times 10^{-3} \text{ (m}^2\text{.V/s)}$ is that of a good conductor.

$$u = \left| \frac{J}{\rho_e} \right| = \left| \frac{\mu_e J}{\sigma} \right| = |\mu_e E| = 1.4 \times 10^{-3} \times (6 \times 10^{-3}) \\ = 8.4 \times 10^{-6} \text{ (m/s)}.$$

P. 4-2 $R_1 = \text{Resistance per unit length of core} = \frac{1}{\sigma S_1} = \frac{1}{\sigma \pi a^2}.$

$R_2 = \text{Resistance per unit length of coating} = \frac{1}{0.1 \sigma S_2}.$

Let $b = \text{Thickness of coating} \rightarrow S_2 = \pi(a+b)^2 - \pi a^2 = \pi(2ab+b^2).$

a) $R_1 = R_2 \rightarrow b = (\sqrt{11} - 1)a = 2.32a.$

b) $I_1 = I_2 = \frac{I}{2}$. $J_1 = \frac{I}{2\pi a^2}$, $J_2 = \frac{I}{2S_2} = \frac{I}{20\pi a^2} = \frac{I}{20\pi a^2}.$

$$E_1 = \frac{J_1}{\sigma} = \frac{I}{2\pi a^2 \sigma}, \quad E_2 = \frac{J_2}{0.1\sigma} = \frac{I}{2\pi a^2 \sigma}.$$

Thus, $J_1 = 10J_2$ and $E_1 = E_2.$

P. 4-3 $\rho_0 = \frac{Q_0}{(4\pi/3)b^3} = \frac{10^{-3}}{(4\pi/3)(0.1)^3} = 0.239 \text{ (C/m}^3\text{)}, \quad \rho = \rho_0 e^{-(\sigma/\epsilon)t}$

a) $R < b$: $\bar{E}_i = \bar{a}_R \frac{(4\pi/3)R^3 \rho}{4\pi \epsilon R^2} = \bar{a}_R \frac{\rho_0 R}{3\epsilon} e^{-(\sigma/\epsilon)t} = \bar{a}_R 7.5 \times 10^9 R e^{-9.42 \times 10^{10} t} \text{ (V/m)}.$

$R > b$: $\bar{E}_o = \bar{a}_R \frac{Q_0}{4\pi \epsilon_0 R^2} = \bar{a}_R \frac{9}{R^2} \times 10^6 \text{ (V/m)}.$

b) $R < b$: $\bar{J}_i = \sigma \bar{E}_i = \bar{a}_R 7.5 \times 10^{10} R e^{-9.42 \times 10^{10} t} \text{ (A/m}^2\text{)}.$

$R > b$: $\bar{J}_o = 0.$

P.4-4 a) $e^{-(\sigma/\epsilon)t} = \frac{\rho}{\rho_0} = 0.01 \rightarrow t = \frac{\ln 100}{(\sigma/\epsilon)} = 4.88 \times 10^{-12} \text{ (s)} = 4.88 \text{ (ps)}$.

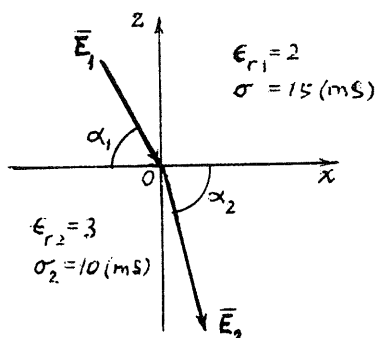
b) $W_i = \frac{\epsilon}{2} \int_{V'} E_i^2 dV' = \frac{2\pi \rho_0 b^3}{45\epsilon} e^{-2(\sigma/\epsilon)t} = (W_i)_0 [e^{-(\sigma/\epsilon)t}]^2$.

$\therefore \frac{W_i}{(W_i)_0} = [e^{-(\sigma/\epsilon)t}]^2 = 0.01^2 = 10^{-4}$ Energy dissipated as heat loss.

c) Electrostatic energy stored outside the sphere $W_o = \frac{\epsilon_0}{2} \int_b^\infty E_o^2 4\pi R^2 dR = \frac{Q_o^2}{8\pi\epsilon_0 b} = 45 \text{ (kJ)}$
— constant.

P.4-5 $I_1 = 0.1 \text{ (A)}, P_{R1} = 3.33 \text{ (mW)}; I_2 = 0.02 \text{ (A)}, P_{R2} = 8.00 \text{ (mW)};$
 $I_3 = 0.0133 \text{ (A)}, P_{R3} = 5.31 \text{ (mW)}; I_4 = 0.0333 \text{ (A)}, P_{R4} = 8.87 \text{ (mW)};$
 $I_5 = 0.0667 \text{ (A)}, P_{R5} = 44.5 \text{ (mW)}. \quad \sum_n P_{Rn} = V_o I_1 = 70 \text{ (mW)}.$
Total resistance seen by the source = $7 \text{ } (\Omega)$.

P.4-6



$\vec{E}_1 = \vec{a}_x 20 - \vec{a}_z 50 \text{ (V/m)}$.

a) $E_{2t} = E_{1t} = 20$.

$J_{2n} = J_{1n} \rightarrow \sigma_2 E_{2n} = \sigma_1 E_{1n}$
 $\rightarrow E_{2n} = \frac{\sigma_1}{\sigma_2} E_{1n} = \frac{15}{10} (-50) = -75$.

$\therefore \vec{E}_2 = \vec{a}_x 20 - \vec{a}_z 75 \text{ (V/m)}$.

b) $\vec{J}_1 = \sigma_1 \vec{E}_1 = 15 \times 10^{-3} (\vec{a}_x 20 - \vec{a}_z 50) = \vec{a}_x 0.3 - \vec{a}_z 0.75 \text{ (A/m}^2\text{)}$.

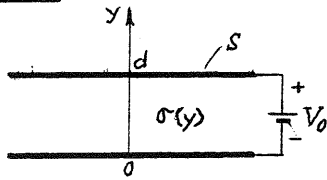
$\vec{J}_2 = \sigma_2 \vec{E}_2 = 10 \times 10^{-3} (\vec{a}_x 20 - \vec{a}_z 75) = \vec{a}_x 0.2 - \vec{a}_z 0.75 \text{ (A/m}^2\text{)}$.

c) $\alpha_1 = \tan^{-1}(\frac{50}{20}) = 68.2^\circ, \quad \alpha_2 = \tan^{-1}(\frac{75}{20}) = 75.1^\circ$.

d) $D_{2n} - D_{1n} = \rho_s \rightarrow \epsilon_2 E_{2n} - \epsilon_1 E_{1n} = \rho_s$.

$\rho_s = \epsilon_0 (-3 \times 75 + 2 \times 50) = -125 \epsilon_0 = -1.105 \text{ (nC/m}^2\text{)}$.

P.4-7



$$\sigma(y) = \sigma_1 + (\sigma_2 - \sigma_1) \frac{y}{d}$$

a) Neglecting fringing effect and assuming a current density:

$$\bar{J} = -\bar{a}_y J_0 \rightarrow \bar{E} = \frac{\bar{J}}{\sigma} = -\bar{a}_y \frac{J_0}{\sigma(y)}$$

$$V_0 = -\int_0^d \bar{E} \cdot \bar{a}_y dy = \int_0^d \frac{J_0 dy}{\sigma_1 + (\sigma_2 - \sigma_1) \frac{y}{d}} = \frac{J_0 d}{\sigma_2 - \sigma_1} \ln \frac{\sigma_2}{\sigma_1}$$

$$R = \frac{V_0}{I} = \frac{V_0}{J_0 s} = \frac{d}{(\sigma_2 - \sigma_1) s} \ln \frac{\sigma_2}{\sigma_1}$$

$$b) (\rho_s)_u = \epsilon_0 E_y(d) = \frac{\epsilon_0 J_0}{\sigma_2} = \frac{\epsilon_0 (\sigma_2 - \sigma_1) V_0}{\sigma_2 d \ln(\sigma_2/\sigma_1)} \quad \text{on upper plate,}$$

$$(\rho_s)_L = -\epsilon_0 E_y(0) = -\frac{\epsilon_0 J_0}{\sigma_1} = -\frac{\epsilon_0 (\sigma_2 - \sigma_1) V_0}{\sigma_1 d \ln(\sigma_2/\sigma_1)} \quad \text{on lower plate.}$$

P.4-8 a) Continuity of the normal component of $\bar{J}$ assures the same current in both media. By Kirchhoff's voltage law:

$$V_0 = (R_1 + R_2) I = \left(\frac{d_1}{\sigma_1 s} + \frac{d_2}{\sigma_2 s} \right) I$$

$$\therefore J = \frac{I}{s} = \frac{V_0}{(d_1/\sigma_1) + (d_2/\sigma_2)} = \frac{\sigma_1 \sigma_2 V_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

b) Two equations are needed for the determination of $\bar{E}_1$ and $\bar{E}_2$:

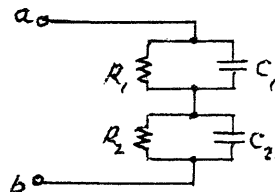
$$V_0 = E_1 d_1 + E_2 d_2$$

$$\text{and } \sigma_1 E_1 = \sigma_2 E_2$$

$$\text{Solving, we have } E_1 = \frac{\sigma_2 V_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

$$\text{and } E_2 = \frac{\sigma_1 V_0}{\sigma_2 d_1 + \sigma_1 d_2}$$

c) Equivalent R-C circuit between terminals a and b:



$$R_1 = \frac{d_1}{\sigma_1 s}$$

$$R_2 = \frac{d_2}{\sigma_2 s}$$

$$C_1 = \frac{\epsilon_1 s}{d_1}$$

$$C_2 = \frac{\epsilon_2 s}{d_2}$$

P. 4-9 a) Same equivalent R-C circuit as that in Problem P. 4-8 with

$$R_1 = \frac{1}{2\pi\sigma_1 L} \ln\left(\frac{c}{a}\right), \quad R_2 = \frac{1}{2\pi\sigma_2 L} \ln\left(\frac{b}{c}\right).$$

$$C_1 = \frac{2\pi\epsilon_1 L}{\ln(c/a)}, \quad C_2 = \frac{2\pi\epsilon_2 L}{\ln(b/c)}.$$

$$b) \quad I = V_0 G = V_0 \frac{1}{R_1 + R_2} = \frac{2\pi\sigma_1\sigma_2 L V_0}{\sigma_1 \ln(b/c) + \sigma_2 \ln(c/a)}.$$

$$J_1 = J_2 = \frac{I}{2\pi r L} = \frac{\sigma_1 \sigma_2 V_0}{r [\sigma_1 \ln(b/c) + \sigma_2 \ln(c/a)]}.$$

P. 4-10 Resistance $R = \frac{\ell}{\sigma s}$. (Eq. 4-16)

— Homogeneous material with a uniform crosssection.

Between top and bottom flat faces: $s = \frac{\pi}{4} (b^2 - a^2)$.

$$R = \frac{4h}{\sigma \pi (b^2 - a^2)}.$$

P. 4-11 Use Laplace's equation in cylindrical coordinates.

$$\nabla^2 V = 0 \longrightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0.$$

Solution: $V(r) = c_1 \ln r + c_2$.

Boundary conditions: $V(a) = V_0$; $V(b) = 0$.

$$\longrightarrow V(r) = V_0 \frac{\ln(b/r)}{\ln(b/a)}.$$

$$\vec{E}(r) = -\vec{a}_r \frac{\partial V}{\partial r} = \vec{a}_r \frac{V_0}{r \ln(b/a)}.$$

$$\vec{J}(r) = \sigma \vec{E}(r).$$

$$I = \int_s \vec{J} \cdot d\vec{s} = \int_0^{\pi/2} \vec{J} \cdot (\vec{a}_r h r d\phi) = \frac{\pi \sigma h V_0}{2 \ln(b/a)}.$$

$$R = \frac{V_0}{I} = \frac{2 \ln(b/a)}{\pi \sigma h}.$$

P. 4-12 Assume a potential difference V_0 between the inner and outer spheres.

$$\nabla^2 V = 0 \rightarrow \frac{1}{R^2} \frac{d}{dR} (R^2 V') = 0. \rightarrow V = \frac{K}{R}. \rightarrow E_R = \frac{K}{R^2}.$$

$$V_0 = -\int_{R_2}^{R_1} E_R dR = -K \int_{R_2}^{R_1} \frac{1}{R^2} dR = K \left(\frac{1}{R_1} - \frac{1}{R_2} \right).$$

$$\rightarrow K = \frac{V_0}{\frac{1}{R_1} - \frac{1}{R_2}}. \quad J_R = \sigma E_R = \frac{\sigma V_0}{\frac{1}{R_1} - \frac{1}{R_2}}.$$

$$I = \int_0^{2\pi} \int_0^\pi J_R R^2 \sin \theta d\theta d\phi = \frac{4\pi \sigma V_0}{\frac{1}{R_1} - \frac{1}{R_2}}.$$

$$R = \frac{V_0}{I} = \frac{1}{4\pi\sigma} \left(\frac{1}{R_1} - \frac{1}{R_2} \right).$$

Chapter 5

Static Magnetic Fields

P. 5-1 $\vec{F} = Q(\vec{E} + \vec{u} \times \vec{B}) = 0.$

$$\begin{aligned}\vec{E} &= -\vec{u} \times \vec{B} = -\vec{a}_x u_0 \times (\vec{a}_x B_x + \vec{a}_y B_y + \vec{a}_z B_z) \\ &= u_0 (\vec{a}_y B_z - \vec{a}_z B_y).\end{aligned}$$

P. 5-2 $\vec{B} = \vec{a}_\phi B_\phi = \vec{a}_\phi \frac{\mu_0 N I}{2\pi r}.$

$$\begin{aligned}\Phi &= \int_s B_\phi ds = \frac{\mu_0 N I}{2\pi} \int_a^b \frac{h}{r} dr \\ &= \frac{\mu_0 N I h}{2\pi} \ln \frac{b}{a}.\end{aligned}$$

If B_ϕ at $r = \frac{a+b}{2}$ is used, $\Phi' = \frac{\mu_0 N I h}{\pi} \left(\frac{b-a}{b+a} \right).$

$$\% \text{ error} = \frac{\Phi' - \Phi}{\Phi} \times 100$$

$$= \left[\frac{2(b-a)}{(b+a) \ln(b/a)} - 1 \right] \times 100.$$

For $\frac{b}{a} = 5$, the error is $\left[\frac{2(5-1)}{(5+1) \ln 5} - 1 \right] \times 100,$
or -17.2% (too low).

P. 5-3 a) Use Eq. (5-32c). $d\vec{L}' = \vec{a}_z dz'$, $\vec{R} = \vec{a}_r r - \vec{a}_z z'.$

$$d\vec{L}' \times \vec{R} = \vec{a}_z dz' \times (\vec{a}_r r - \vec{a}_z z') = \vec{a}_\phi r dz'.$$

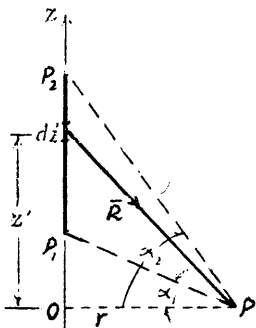
$$\vec{B}_p = \vec{a}_\phi \frac{\mu_0 I}{4\pi} \int \frac{r dz'}{(z'^2 + r^2)^{3/2}}.$$

Let $z' = r \tan \alpha$, $dz' = r \sec^2 \alpha d\alpha.$

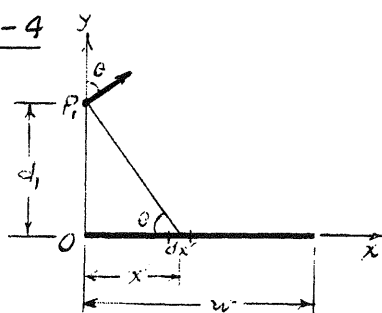
$$\begin{aligned}\vec{B}_p &= \vec{a}_\phi \frac{\mu_0 I}{4\pi r} \int_{\alpha_1}^{\alpha_2} \cos \alpha d\alpha \\ &= \vec{a}_\phi \frac{\mu_0 I}{4\pi r} (\sin \alpha_2 - \sin \alpha_1).\end{aligned}$$

b) For an infinitely long wire: $\alpha_2 \rightarrow 90^\circ$ and $\alpha_1 \rightarrow -90^\circ$

$\vec{B}_p$ becomes $\vec{a}_\phi \frac{\mu_0 I}{2\pi r}$, as in Eq. (5-35).



P.5-4



Use Eq. (5-35):

$$d\vec{B}_{P_1} = \vec{a}_x dB_x + \vec{a}_y dB_y$$

$$= \vec{a}_x (dB_{P_1}) \sin \theta + \vec{a}_y (dB_{P_1}) \cos \theta,$$

$$\text{where } dB_{P_1} = \frac{\mu_0 (I/w) dx'}{2\pi (x'^2 + d_1^2)^{3/2}},$$

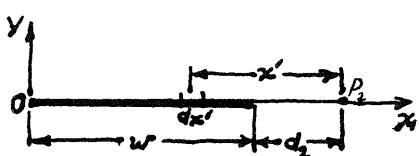
$$\sin \theta = \frac{d}{(x'^2 + d^2)^{1/2}}, \quad \cos \theta = \frac{x'}{(x'^2 + d^2)^{1/2}}.$$

$$\vec{B}_{P_1} = \vec{a}_x B_x + \vec{a}_y B_y,$$

$$\text{where } B_x = \frac{\mu_0 I d}{2\pi w} \int_0^w \frac{dx'}{x'^2 + d^2} = \frac{\mu_0 I}{2\pi w} \tan^{-1} \left(\frac{w}{d} \right),$$

$$\text{and } B_y = \frac{\mu_0 I}{2\pi w} \int_0^w \frac{x' dx'}{x'^2 + d^2} = \frac{\mu_0 I}{4\pi w} \ln \left(1 + \frac{w}{d} \right).$$

P.5-5



I flows into the paper
(in $-\vec{a}_z$ direction).

$$d\vec{B}_{P_2} = -\vec{a}_z \frac{\mu_0 I dx'}{2\pi w x'}.$$

$$\vec{B}_{P_2} = -\vec{a}_z \frac{\mu_0 I}{2\pi w} \int_{d_2}^{d_2+w} \frac{dx'}{x'} = -\vec{a}_z \frac{\mu_0 I}{2\pi w} \ln \left(1 + \frac{w}{d_2} \right).$$

P.5-6 Apply Ampère's circuital law, Eq. (5-10), and assume the

medium to be nonmagnetic:

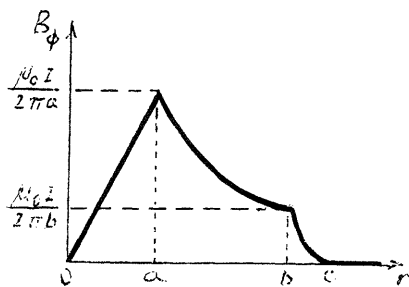
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I.$$

$$\text{For } 0 \leq r \leq a, \quad \vec{B} = \vec{a}_\phi \frac{\mu_0 r I}{2\pi a^2}.$$

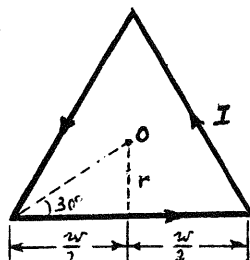
$$\text{For } a \leq r \leq b, \quad \vec{B} = \vec{a}_\phi \frac{\mu_0 I}{2\pi r}.$$

$$\text{For } b \leq r \leq c, \quad \vec{B} = \vec{a}_\phi \left(\frac{c^2 - r^2}{c^2 - b^2} \right) \frac{\mu_0 I}{2\pi r}.$$

$$\text{For } r \geq c, \quad \vec{B} = 0.$$



P. 5-7



Assume that the current flows in the counterclockwise direction in a triangle lying in the xy -plane.

From Eq. (5-34) and noting that

$$L = \frac{w}{2} \text{ and } r = \frac{w}{2} \tan 30^\circ = \frac{w}{2\sqrt{3}},$$

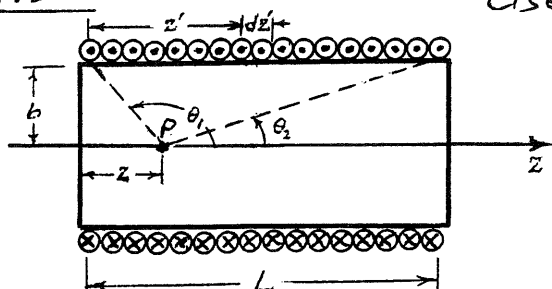
we have

$$\vec{B} = 3 \left(\vec{a}_z \frac{\mu_0 I L}{2\pi r \sqrt{L^2 + r^2}} \right) \text{ at } O.$$

$$L/r = \sqrt{3}, \quad \sqrt{L^2 + r^2} = \frac{w}{\sqrt{3}}.$$

$$\vec{B} = \vec{a}_z \frac{3\mu_0 I}{2\pi} \frac{\sqrt{3}}{w/\sqrt{3}} = \vec{a}_z \frac{9\mu_0 I}{2\pi w}.$$

P. 5-8



Use Eq. (5-37):

$$d\vec{B} = \vec{a}_z \frac{\mu I b^2}{2[(z'-z)^2 + b^2]^{3/2}} \left(\frac{N}{L}\right) dz',$$

$$\vec{B} = \vec{a}_z \frac{\mu N I b^2}{2L}$$

$$= \vec{a}_z \frac{\mu N I}{2L} \left[\frac{L-z}{\sqrt{(L-z)^2 + b^2}} - \frac{z}{\sqrt{z^2 + b^2}} \right]$$

For $L \rightarrow \infty$, $\theta_2 \rightarrow 0$ and $\theta_1 \rightarrow \pi$,

and $\vec{B} \rightarrow \vec{a}_z \frac{\mu N I}{L} = \vec{a}_z \mu n I$.

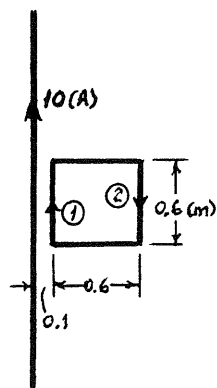
$$= \vec{a}_z \frac{\mu N I}{2L} (\cos \theta_2 - \cos \theta_1).$$

P. 5-9 a) $\vec{B} = \nabla \times \vec{A} = \vec{a}_\phi \frac{\mu_0 I}{2\pi r} = -\vec{a}_\phi \frac{\partial A_z}{\partial r}$. (No change with z .)

$$\frac{dA_z}{dr} = -\frac{\mu_0 I}{2\pi r} \rightarrow A_z = -\frac{\mu_0 I}{2\pi} \ln r + c.$$

$$A_z = 0 \text{ at } r = r_0 \rightarrow c = \frac{\mu_0 I}{2\pi} \ln r_0.$$

$$\therefore \vec{A} = \vec{a}_z A_z = \vec{a}_z \frac{\mu_0 I}{2\pi} \ln \left(\frac{r_0}{r} \right).$$



b) Use $\Phi = \oint \vec{A} \cdot d\vec{\ell}$.

Horizontal sides have no effect.

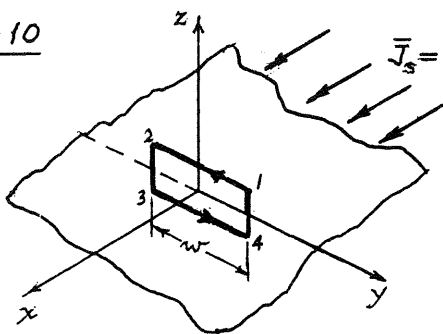
Side ①: $\int \vec{A} \cdot d\vec{\ell} = \left(\frac{\mu_0 I}{2\pi} \ln \frac{r_o}{0.1} \right) \times 0.6$.

Side ②: $\int \vec{A} \cdot d\vec{\ell} = - \left(\frac{\mu_0 I}{2\pi} \ln \frac{r_o}{0.7} \right) \times 0.6$.

$$\begin{aligned} \oint \vec{A} \cdot d\vec{\ell} &= \left(\frac{\mu_0 I}{2\pi} \ln \frac{0.7}{0.1} \right) \times 0.6 \\ &= \frac{(4\pi \times 10^{-7}) \times 10 \times 0.6}{2\pi} \ln 7 \\ &= 2.34 \times 10^{-6} \text{ (Wb)}. \end{aligned}$$

$\therefore \Phi = 2.34 \text{ (}\mu\text{Wb)}.$

P.5-10



Infinite current sheet

$\rightarrow \vec{B}$ antisymmetrical and independent of x and y .

a) Apply Ampère's circuital law to path 12341:

$$\oint_c \vec{B} \cdot d\vec{\ell} = \mu_0 I \rightarrow 2wB_y = \mu_0 J_{s0} w$$

$$\rightarrow B_y = \begin{cases} -\mu_0 J_{s0}/2 & \text{at } (0,0,z), \\ +\mu_0 J_{s0}/2 & \text{at } (0,0,-z). \end{cases}$$

or, $\vec{B} = \frac{\mu_0}{2} \vec{J}_s \times \vec{a}_n$.

b) For $z > 0$, $\vec{\nabla} \times \vec{A} = \vec{B} = \vec{a}_y \left(-\frac{\mu_0 J_{s0}}{2} \right)$.

$\vec{A}$ is independent of x and y .

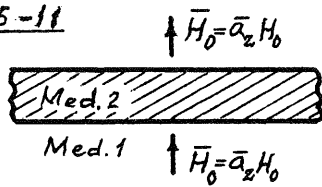
$$\frac{dA_x}{dz} = -\frac{\mu_0 J_{s0}}{2}.$$

$$A_x = -\frac{\mu_0 J_{s0}}{2} z + c.$$

At $z = z_0$, $A_x = 0 = -\frac{\mu_0 J_{s0}}{2} z_0 + c \rightarrow c = \frac{\mu_0 J_{s0}}{2} z_0$.

$$\therefore \vec{A} = -\frac{\mu_0}{2} (z - z_0) \vec{J}_s.$$

P. 5-11



a) Given $\vec{B}_2 = \mu_2 \vec{H}_2$.

$$B_{2z} = B_{1z} \rightarrow \mu_2 H_2 = \mu_0 H_0 \rightarrow \vec{H}_2 = \vec{a}_z H_2 = \vec{a}_z \frac{\mu_0}{\mu_2} H_0.$$

b) Given $\vec{B}_2 = \mu_0 (\vec{H}_2 + \vec{M}_i)$.

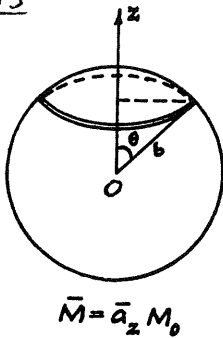
$$B_{2z} = B_{1z} \rightarrow \mu_0 (H_2 + M_i) = \mu_0 H_0 \rightarrow \vec{H}_2 = \vec{a}_z (H_0 - M_i).$$

P. 5-12

a) $r < a$: $\left. \begin{aligned} \vec{H} &= \vec{a}_z n I, \\ \vec{B} &= \vec{a}_z \mu n I, \\ \vec{M} &= \frac{\vec{B}}{\mu_0} - \vec{H} = \vec{a}_z \left(\frac{\mu}{\mu_0} - 1 \right) n I. \end{aligned} \right\} \text{Eq. (6-14).}$ $a < r < b$: $\left. \begin{aligned} \vec{H} &= \vec{a}_z n I, \\ \vec{B} &= \vec{a}_z \mu_0 n I, \\ \vec{M} &= 0. \end{aligned} \right\}$

b) $\vec{J}_m = \nabla \times \vec{M} = 0$; $\vec{J}_{ms} = \vec{M} \times \vec{a}_n = (\vec{a}_z \times \vec{a}_r) \left(\frac{\mu}{\mu_0} - 1 \right) n I = \vec{a}_\phi \left(\frac{\mu}{\mu_0} - 1 \right) n I.$

P. 5-13



a) $\vec{J}_m = \nabla \times \vec{M} = 0.$

$$\vec{J}_{ms} = (\vec{a}_R \cos \theta - \vec{a}_\theta \sin \theta) M \times \vec{a}_R = \vec{a}_\phi M_0 \sin \theta.$$

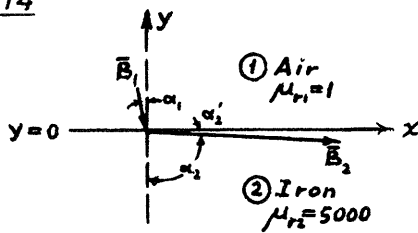
b) Apply Eq. (5-37) to a loop of radius $b \sin \theta$ carrying a current $J_{ms} b d\theta$:

$$d\vec{B} = \vec{a}_z \frac{\mu_0 (J_{ms} b d\theta) (b \sin \theta)^2}{2 (b^2)^{3/2}} = \vec{a}_z \frac{\mu_0 M_0}{2} \sin^3 \theta.$$

$$\vec{B} = \int d\vec{B} = \vec{a}_z \frac{\mu_0 M_0}{2} \int_0^\pi \sin^3 \theta d\theta = \vec{a}_z \frac{2}{3} \mu_0 M_0 = \frac{2}{3} \mu_0 \vec{M},$$

at the center O.

P. 5-14



a) $\vec{B}_1 = \vec{a}_x 2 - \vec{a}_y 10 \text{ (mT)},$

$$\vec{B}_2 = \vec{a}_x B_{2x} - \vec{a}_y B_{2y}.$$

$$H_{2x} = \frac{B_{2x}}{5000 \mu_0} = H_{1x} = \frac{2}{\mu_0} \rightarrow B_{2x} = 10,000 \text{ (mT)},$$

$$B_{2y} = B_{1y} = -10 \text{ (mT)}.$$

$$\therefore \vec{B}_2 = \vec{a}_x 10,000 - \vec{a}_y 10 \text{ (mT)}.$$

$$\tan \alpha_2 = \frac{\mu_2}{\mu_1} \tan \alpha_1 = 5000 \left(\frac{B_{1x}}{B_{1y}} \right) = 1,000 \rightarrow \alpha_2 = 89.94^\circ, \alpha_2' = 0.04^\circ.$$

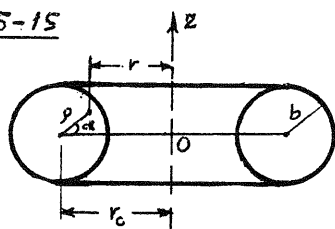
b) If $\vec{B}_2 = \bar{a}_x 10 + \bar{a}_y$ (mT), $\vec{B}_1 = \bar{a}_x B_{1x} + \bar{a}_y B_{1y}$.

$$H_{1x} = \frac{B_{1x}}{\mu_1} = H_{2x} = \frac{B_{2x}}{\mu_2} \rightarrow B_{1x} = \frac{1}{\mu_{r2}} B_{2x} = \frac{10}{5000} = 0.002.$$

$$B_{1y} = B_{2y} = 2. \quad \therefore \vec{B}_1 = \bar{a}_x 0.002 + \bar{a}_y 2 \text{ (mT)}.$$

$$\alpha_1 = \tan^{-1} \frac{B_{1x}}{B_{1y}} \approx \frac{0.002}{2} = 0.001 \text{ (rad)} = 0.057^\circ$$

P. 5-15



$$\vec{B} = \bar{a}_\phi B_\phi = \bar{a}_\phi \frac{\mu_0 N I}{2\pi r}, \quad r = r_0 - \rho \cos \alpha.$$

$$\Phi = \frac{\mu_0 N I}{2\pi} \int_0^b \int_0^{2\pi} \frac{\rho d\alpha d\rho}{r_0 - \rho \cos \alpha} = \mu_0 N I (r_0 - \sqrt{r_0^2 - b^2}).$$

$$\therefore L = \frac{N\Phi}{I} = \mu_0 N^2 (r_0 - \sqrt{r_0^2 - b^2}).$$

$$\text{If } r_0 \gg b, \quad B_\phi \approx \frac{\mu_0 N I}{2\pi r_0} \text{ (constant)}.$$

$$\Phi \approx B_\phi S = B_\phi (\pi b^2) = \frac{\mu_0 N b^2 I}{2 r_0} \rightarrow L \approx \frac{\mu_0 N^2 b^2}{2 r_0}.$$

P. 5-16 For I in the long straight wire, $\vec{B} = \bar{a}_\phi \frac{\mu_0 I}{2\pi r}$.

$$\begin{aligned} \Lambda_{12} &= \int_S \vec{B} \cdot d\vec{s} = \int B_\phi \frac{2}{\sqrt{3}} (r-d) dr = \frac{\mu_0 I}{\pi \sqrt{3}} \int_d^{d+\frac{\sqrt{3}}{2}b} \left(\frac{r-d}{r} \right) dr \\ &= \frac{\mu_0 I}{\pi \sqrt{3}} \left[\frac{\sqrt{3}}{2} b - d \ln \left(1 + \frac{\sqrt{3}b}{2d} \right) \right], \end{aligned}$$

$$L_{12} = \frac{\Lambda_{12}}{I} = \frac{\mu_0}{\pi} \left[\frac{b}{2} - \frac{d}{\sqrt{3}} \ln \left(1 + \frac{\sqrt{3}b}{2d} \right) \right].$$

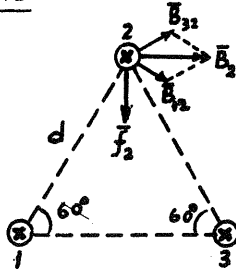
P. 5-17 Approximate the magnetic flux due to the long loop linking with the small loop by that due to two infinitely long wires carrying equal and opposite current I .

$$\Lambda_{12} = \frac{\mu_0 h_2 I}{2\pi} \int_0^{w_2} \left(\frac{1}{d+x} - \frac{1}{w_1+d+x} \right) dx$$

$$= \frac{\mu_0 h_2 I}{2\pi} \ln \left(\frac{w_2+d}{d} \cdot \frac{w_1+d}{w_1+w_2+d} \right).$$

$$L_{12} = \frac{\Lambda_{12}}{I} = \frac{\mu_0 h_2}{2\pi} \ln \frac{(w_1+d)(w_2+d)}{d(w_1+w_2+d)}.$$

P. 5-18



$$I_1 = I_2 = I_3 = 25 \text{ (A)}; \quad d = 0.15 \text{ (m)}.$$

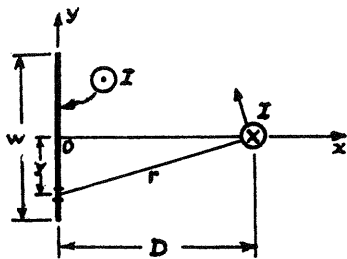
$$\vec{B}_2 = \vec{a}_x 2B_{12} \cos 30^\circ = \vec{a}_x \frac{\sqrt{3} \mu_0 I}{2\pi d}.$$

Force per unit length on wire 2:

$$\begin{aligned} \vec{f}_2 &= -\vec{a}_y I B_2 = -\vec{a}_y \frac{\sqrt{3} \mu_0 I^2}{2\pi d} \\ &= -\vec{a}_y 1150 \mu_0 = -\vec{a}_y 1.44 \times 10^{-3} \text{ (N/m)}. \end{aligned}$$

Forces on all three wires are of equal magnitude and toward the center of the triangle.

P. 5-19 Magnetic field intensity at the wire due to the current $dI = \frac{I}{w} dy$ in an elemental dy is



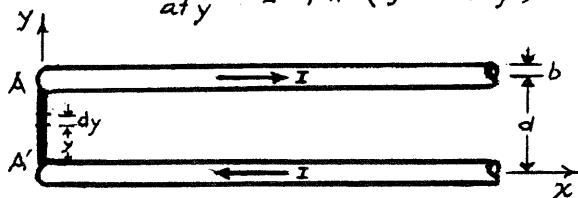
$$|d\vec{H}| = \frac{dI}{2\pi r} = \frac{I dy}{2\pi w \sqrt{D^2 + y^2}}.$$

Symmetry $\rightarrow \vec{H}$ at the wire has only a y -component.

$$\begin{aligned} \vec{H} &= \vec{a}_y \int (dH) \cdot \left(\frac{D}{r}\right) = \vec{a}_y 2 \int_0^{w/2} \frac{ID dy}{2\pi w (D^2 + y^2)} \\ &= \vec{a}_y \frac{I}{\pi w} \tan^{-1} \left(\frac{w}{2D} \right). \end{aligned}$$

$$\vec{f}' = \vec{I} \times \vec{B} = (-\vec{a}_x I) \times (\mu_0 \vec{H}) = \vec{a}_x \frac{\mu_0 I^2}{\pi w} \tan^{-1} \left(\frac{w}{2D} \right) \text{ (N/m)}.$$

P. 5-20 $\vec{B} = -\vec{a}_z \frac{\mu_0 I}{4\pi} \left(\frac{1}{y} + \frac{1}{d-y} \right)$, $d\vec{\ell} = \vec{a}_y dy$.



(A rail-gun problem.)

$$d\vec{F} = I d\vec{\ell} \times \vec{B}$$

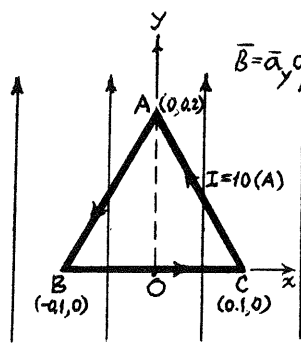
$$= -\vec{a}_x \frac{\mu_0 I^2}{4\pi} \left(\frac{1}{y} + \frac{1}{d-y} \right) dy$$

$$\vec{F} = -\vec{a}_x \frac{\mu_0 I^2}{4\pi} \int_b^{d-b} \left(\frac{1}{y} + \frac{1}{d-y} \right) dy$$

$$= -\vec{a}_x \frac{\mu_0 I^2}{2\pi} \ln \left(\frac{d}{b} - 1 \right).$$

P.5-21 Force in a uniform magnetic field:

$$\vec{F} = I \vec{L} \times \vec{B} = -\vec{B} \times (I \vec{L}).$$



$\vec{B} = \vec{a}_y 0.5 \text{ (T)}$
 $-\vec{B} \times I(\vec{AB}) \quad I(\vec{CA}) \quad I(\vec{BC})$
 $-(\vec{a}_y 0.5) \times 10(-\vec{a}_x 0.1 - \vec{a}_y 0.2), 10(\vec{a}_x 0.1 + \vec{a}_y 0.2), 10 \vec{a}_x 0.2$
Force: $-\vec{a}_x 0.5 \quad -\vec{a}_x 0.5 \quad \vec{a}_x 1.0 \text{ (N)}$
Torque on loop:
 $\vec{T} = \vec{m} \times \vec{B} = (\vec{a}_z I S) \times \vec{B}$
 $= (\vec{a}_z 10 \times \frac{1}{2} \times 0.2 \times 0.2) \times (\vec{a}_y 0.5) = -\vec{a}_x 0.1 \text{ (N.m)}.$

P.5-22 $\vec{B}_2$ at the center of the large circular turn of wire carrying a current I_2 is (by setting $z = 0$ in Eq. 5-37):

$$\vec{B}_2 = \vec{a}_{n2} \frac{\mu_0 I_2}{2 r_2}.$$

Torque on the small circular turn of wire carrying a current I_1 is

$$\begin{aligned} \vec{T} &= \vec{m}_1 \times \vec{B}_2 \cong (\vec{a}_{n1} I_1 \pi r_1^2) \times (\vec{a}_{n2} \frac{\mu_0 I_2}{2 r_2}) \\ &= (\vec{a}_{n1} \times \vec{a}_{n2}) \frac{\mu_0 I_1 I_2 \pi r_1^2}{2 r_2}, \end{aligned}$$

which is a torque having a magnitude $\frac{\mu_0 I_1 I_2 \pi r_1^2}{2 r_2} \sin \theta$ and a direction tending to align the magnetic fluxes produced by I_1 and I_2 .

Chapter 6

Time-Varying Fields and Maxwell's Equations

P. 6-1 $\mathcal{V} = - \int_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{s}$
 $= - \int_S \frac{\partial}{\partial t} (\bar{\nabla} \times \bar{A}) \cdot d\bar{s}$
 $= - \oint \frac{\partial \bar{A}}{\partial t} \cdot d\bar{\ell}$

P. 6-2 $\bar{B} = \bar{a}_z 3 \cos(5\pi 10^7 t - \frac{1}{3}\pi y) \times 10^{-6} \text{ (T)}$
 $\int_S \bar{B} \cdot d\bar{s} = \int_0^{0.3} \bar{a}_z 3 \cos(5\pi 10^7 t - \frac{1}{3}\pi y) 10^{-6} (\bar{a}_z 0.1 dy)$
 $= -\frac{0.9}{\pi} [\sin(5\pi 10^7 t - 0.1\pi) - \sin 5\pi 10^7 t] \times 10^{-6} \text{ (Wb)}$
 $\mathcal{V} = - \frac{d}{dt} \int_S \bar{B} \cdot d\bar{s} = 4.5 [\cos(5\pi 10^7 t - 0.1\pi) - \cos 5\pi 10^7 t] \text{ (V)}$
 $i = \frac{\mathcal{V}}{2R} = 0.15 [\cos(5\pi 10^7 t - 0.1\pi) - \cos 5\pi 10^7 t]$
 $= 0.023 \sin(5\pi 10^7 t - 0.05\pi) \text{ (A)}$
 $= 23 \sin(5\pi 10^7 t - 9^\circ) \text{ (mA)}$

P. 6-3 Using phasors with a sine reference:

$$\bar{B}_1 = \bar{a}_\phi \frac{\mu_0 I_1}{2\pi r} \rightarrow \Phi_{12} = \int_{S_2} \bar{B}_1 \cdot d\bar{s}_2 = \frac{\mu_0 I_1 h}{2\pi} \int_d^{d+w} \frac{dr}{r}$$

$$V_2 = - \frac{d\Phi_{12}}{dt} \rightarrow \text{Phasors: } V_2 = -j\omega \Phi_{12} = \frac{\mu_0 I_1 h}{2\pi} \ln\left(1 + \frac{w}{d}\right)$$

$$I_2 = \frac{V_2}{R + j\omega L} = - \frac{j\omega \mu_0 I_1 h}{2\pi(R + j\omega L)} \ln\left(1 + \frac{w}{d}\right)$$

$$= - \frac{\omega \mu_0 I_1 h}{2\pi(\omega L - jR)} \ln\left(1 + \frac{w}{d}\right) = - \frac{\omega \mu_0 I_1 h}{2\pi \sqrt{R^2 + \omega^2 L^2}} \ln\left(1 + \frac{w}{d}\right) e^{j \tan^{-1}(R/\omega L)}$$

$$\rightarrow i_2 = - \frac{\omega \mu_0 I_1 h}{2\pi \sqrt{R^2 + \omega^2 L^2}} \ln\left(1 + \frac{w}{d}\right) \sin\left(\omega t + \tan^{-1} \frac{R}{\omega L}\right)$$

P. 6-4 $\vec{B}_1 = -\vec{a}_x \frac{\mu_0 I_0}{2\pi r}$.

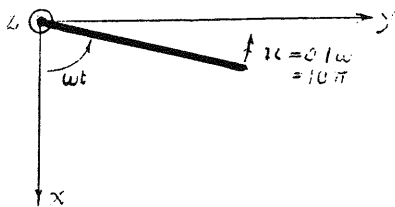
Induced emf in loop $= \oint (\vec{u}_2 \times \vec{B}_1) \cdot d\vec{\ell}_2$.

$\rightarrow \mathcal{V}_2 = \frac{\mu_0 I_0 h u_0}{2\pi} \left(\frac{1}{d} - \frac{1}{d+w} \right),$

in a clockwise direction.

$i_2 = -\frac{\mathcal{V}_2}{R} = -\frac{\mu_0 I_0 h u_0 w}{2\pi d(d+w)}$.

P. 6-5



a) If $L = 0$:

$$\begin{aligned} i &= \frac{1}{R} (\vec{u} \times \vec{B}) \cdot (-\vec{a}_z 0.1) \\ &= \frac{1}{0.5} (10\pi \times 0.04) \times 0.1 \sin \omega t \\ &= 0.251 \sin 100\pi t \text{ (A)}. \end{aligned}$$

b) If $L = 0.0035$ (H):

$$\begin{aligned} \omega L &= 100\pi \times 0.0035 = 1.1 \text{ } (\Omega), \\ \frac{1}{R + j\omega L} &= \frac{1}{0.5 + j1.1} = \frac{1}{1.208 / 65.6^\circ}. \end{aligned}$$

$$\begin{aligned} \rightarrow i_2 &= \frac{(10\pi \times 0.04) \times 0.1}{1.208} \sin(\omega t - 65.6^\circ) \\ &= 0.104 \sin(100\pi t - 65.6^\circ) \text{ (A)}. \end{aligned}$$

P. 6-6

$$\begin{aligned} \Phi &= \vec{B}(t) \cdot \vec{S}(t) = -5 \cos \omega t \times 0.2 (0.7 - x) \\ &= -\cos \omega t [0.7 - 0.35(1 - \cos \omega t)] \\ &= -0.35 \cos \omega t (1 + \cos \omega t) \text{ (mT)}. \end{aligned}$$

$$\begin{aligned} i &= -\frac{1}{R} \frac{d\Phi}{dt} = -\frac{1}{R} 0.35 \omega (\sin \omega t + \sin 2\omega t) \\ &= -1.75 \omega (\sin \omega t + \sin 2\omega t) \\ &= -1.75 \omega \sin \omega t (1 + 2 \cos \omega t) \text{ (mA)}. \end{aligned}$$

P. 6-7 Conduction current density: σE .

Displacement current density: $j\omega D = j\omega\epsilon_0\epsilon_r E$.

For equal magnitude: $\sigma = 2\pi\epsilon_0\epsilon_r f$,

$$\text{or } f = \frac{\sigma}{2\pi(\epsilon_0\epsilon_r)} = 18 \times 10^9 \left(\frac{\sigma}{\epsilon_r} \right) \text{ (Hz)}.$$

a) Seawater: $f = 18 \times 10^9 \left(\frac{4}{72} \right) = 10^9 \text{ (Hz)} = 1 \text{ (GHz)}$.

b) Moist soil: $f = 18 \times 10^9 \left(\frac{10^{-3}}{2.5} \right) = 7.2 \times 10^6 \text{ (Hz)}$,
or 7.2 (MHz).

P. 6-8

a) $\left| \frac{\text{Displacement current}}{\text{Conduction current}} \right| = \frac{\omega\epsilon}{\sigma} = \frac{(2\pi \times 100 \times 10^9) \times \frac{1}{36\pi} \times 10^{-9}}{5.70 \times 10^7}$
 $= 9.75 \times 10^{-8}$.

b) In a source-free conductor:

$$\nabla \times \vec{H} = \sigma \vec{E}, \quad (1)$$

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}. \quad (2)$$

$$\nabla \times (1): \nabla \times \nabla \times \vec{H} = \nabla(\nabla \cdot \vec{H}) - \nabla^2 \vec{H} = \sigma \nabla \times \vec{E}. \quad (3)$$

But $\nabla \cdot \vec{H} = 0$, Eq. (3) becomes

$$\nabla^2 \vec{H} + \sigma \nabla \times \vec{E} = 0. \quad (4)$$

Combining (2) and (4):

$$\nabla^2 \vec{H} - j\omega\mu\sigma\vec{H} = 0.$$

P. 6-9 $\vec{H}_1 = \vec{a}_x 30 + \vec{a}_y 40 + \vec{a}_z 20$. $B_{2n} = B_{1n} \rightarrow H_{2z} = \frac{1}{\mu_{r2}} H_{1z} = 10$.

$$\vec{a}_{n2} \times (\vec{H}_1 - \vec{H}_2) = \vec{J} \quad \rightarrow B_{2z} = B_{1z} = 20\mu_0.$$

$$\rightarrow \vec{a}_z \times (\vec{a}_x 30 + \vec{a}_y 40 - \vec{H}_2) = \vec{a}_z 5.$$

$$\rightarrow H_{2x} = 30, H_{2y} = 45.$$

a) $\vec{H}_2 = \vec{a}_x 30 + \vec{a}_y 45 + \vec{a}_z 10 \text{ (A/m)}$. b) $\vec{B}_2 = 2\mu_0 \vec{H}_2 \text{ (T)}$.

c) $\alpha_1 = \tan^{-1} \frac{\sqrt{30^2 + 40^2}}{20} = 68.2^\circ$. d) $\alpha_2 = \tan^{-1} \frac{\sqrt{30^2 + 45^2}}{10} = 79.5^\circ$.

P. 6-10 Medium 1: Free space.

Medium 2: $\mu_2 \rightarrow \infty$. H_2 must be zero so that B_2 is not infinite.

Boundary Conditions: $\vec{a}_{n2} \times \vec{H}_1 = \vec{J}_s$, $B_{1n} = B_{2n}$.

$$E_{1t} = E_{2t}, \quad \vec{a}_{n2} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s.$$

P. 6-13 $E(z, t) = 50 \cos(2\pi 10^9 t - kz)$ (V/m) in air.

$$E_0 = 50 \text{ (V/m)}.$$

$$f = 10^9 \text{ (Hz)}, \quad T = \frac{1}{f} = 10^{-9} \text{ (s)}.$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^9} = 0.3 \text{ (m)},$$

$$k = \frac{2\pi}{\lambda} = \frac{20}{3} \pi.$$

a) At $z = 100.125\lambda$, $kz = 200.25\pi$, which is same as for $kz = 0.25\pi$, or $\pi/4$.

It is a plot of $E(t) = 50 \cos 2\pi 10^9 (t - T/8)$ (V/m).

b) At $z = -100.125\lambda$, it is a plot of

$$E(t) = 50 \cos 2\pi 10^9 (t + T/8) \text{ (V/m)}.$$

c) At $t = T/4$, we plot versus z the following sinusoidal function:

$$\begin{aligned} E(z, \frac{T}{4}) &= 50 \cos(-kz + \frac{\omega T}{4}) = 50 \cos[-k(z - \frac{\lambda}{4})] \\ &= 50 \cos \frac{20\pi}{3} (z - 0.075) \text{ (V/m)}. \end{aligned}$$

P. 6-14 Use phasors and cosine reference.

$$\vec{E} = \vec{a}_x E_0 e^{j\psi}; \quad \vec{E}_1 = \vec{a}_x 0.03 e^{-j\pi/2}; \quad \vec{E}_2 = \vec{a}_x 0.04 e^{-j\pi/3}$$

$$\begin{aligned} \vec{E} &= \vec{E}_1 + \vec{E}_2 = \vec{a}_x [0.03 e^{-j\pi/2} + 0.04 e^{-j\pi/3}] \\ &= \vec{a}_x [-j0.03 + 0.04(\frac{1}{2} - j\frac{\sqrt{3}}{2})] = \vec{a}_x 0.068 e^{-j72.8^\circ} \end{aligned}$$

$$\longrightarrow E_0 = 0.068 \text{ (V/m)}, \quad \psi = -72.8^\circ.$$

P. 6-15 Use phasors and cosine reference.

$$\bar{H} = \bar{a}_\phi H_0, \quad \bar{H}_1 = \bar{a}_\phi 10^{-4} e^{-j\pi/2}, \quad \bar{H}_2 = \bar{a}_\phi 2 \times 10^{-4} e^{j\alpha}$$

$$\begin{aligned} \rightarrow H_0 &= 10^{-4}(-j + 2e^{j\alpha}) \\ &= 10^{-4} [2\cos\alpha + j(2\sin\alpha - 1)] \end{aligned}$$

$$2\sin\alpha - 1 = 0 \rightarrow \alpha = 30^\circ, \text{ or } \pi/6 \text{ (rad.)}$$

$$H_0 = 2 \times 10^{-4} \cos 30^\circ = 1.73 \times 10^{-4} \text{ (A/m).}$$

P. 6-17 See Section 10-2, pp. 428-429, Eqs. (10-6) and (10-7).

P. 6-18 a) $k = \frac{\omega}{c} = \frac{2\pi(60 \times 10^6)}{3 \times 10^8} = 0.4\pi \text{ (rad/m).}$

b)
$$\begin{aligned} \bar{H} &= \frac{1}{-j\omega\mu_0} \bar{\nabla} \times \bar{E} = \frac{j}{\omega\mu_0} \begin{vmatrix} \bar{a}_r & \bar{a}_\phi & \bar{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ E_r & 0 & 0 \end{vmatrix} \\ &= \frac{j}{\omega\mu_0} \bar{a}_\phi \frac{\partial E_r}{\partial z} = \bar{a}_\phi \frac{j}{\omega\mu_0} (-j k) \frac{E_0}{r} e^{-jkz} \\ &= \bar{a}_\phi \frac{k}{\omega\mu_0} \frac{E_0}{r} e^{-jkz} = \bar{a}_\phi \frac{E_0}{120\pi r} e^{-j0.4\pi z} \text{ (A/m).} \end{aligned}$$

c) $\bar{J}_s \Big|_{r=a} = \bar{a}_z H_\phi \Big|_{r=a} = \bar{a}_z \frac{E_0}{120\pi a} e^{-j0.4\pi z} \text{ (A/m).}$

$\bar{J}_s \Big|_{r=b} = -\bar{a}_z H_\phi \Big|_{r=b} = -\bar{a}_z \frac{E_0}{120\pi b} e^{-j0.4\pi z} \text{ (A/m).}$

P.6-19 $k = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \text{ (rad/m)}.$

$$\vec{H} = \frac{j}{\omega\mu_0} \vec{\nabla} \times \vec{E}.$$

In phasor form: $\vec{E} = \bar{a}_\theta \frac{10^{-3}}{R} \sin\theta e^{jkR}.$

From Eq. (2-99):
$$\vec{H} = \frac{j}{\omega\mu_0 R^2 \sin\theta} \begin{vmatrix} \bar{a}_r & \bar{a}_\theta R & \bar{a}_\phi R \sin\theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ 0 & 10^{-3} \sin\theta e^{jkR} & 0 \end{vmatrix}$$

$$= \bar{a}_\phi \frac{10^{-3}}{R} \sin\theta e^{jkR}$$

In instantaneous form:

$$\vec{H}(R, \theta; t) = \bar{a}_\phi \frac{10^{-3}}{120\pi R} \sin\theta \cos(2\pi \cdot 10^9 t - 20\pi R/3) \text{ (A/m)}.$$

P.6-20 In phasor form:

$$\vec{E} = \bar{a}_y 0.1 \sin(10\pi x) e^{-j\beta z}. \quad \textcircled{1}$$

$$\vec{H} = -\frac{1}{j\omega\mu_0} \vec{\nabla} \times \vec{E}$$

$$= \frac{j}{\omega\mu_0} \left[\bar{a}_x j0.1\beta \sin(10\pi x) + \bar{a}_z 0.1(10\pi) \cos(10\pi x) \right] e^{-j\beta z} \quad \textcircled{2}$$

$$\vec{E} = \frac{1}{j\omega\epsilon_0} \vec{\nabla} \times \vec{H}$$

$$= \bar{a}_y \frac{0.1}{\omega^2\mu_0\epsilon_0} \left[(10\pi)^2 + \beta^2 \right] \sin(10\pi x) e^{-j\beta z} \quad \textcircled{3}$$

Equating ① and ③: $(10\pi)^2 + \beta^2 = \omega^2\mu_0\epsilon_0 = 400\pi^2$

$$\longrightarrow \beta = \sqrt{300}\pi = 54.4 \text{ (rad/m)}.$$

From ②: $\vec{H}(x, z; t) = \Re \{ \vec{H} e^{j\omega t} \}$

$$= -\bar{a}_x 2.30 \times 10^{-4} \sin(10\pi x) \cos(6\pi \cdot 10^9 t - 54.4 z)$$

$$- \bar{a}_z 1.33 \times 10^{-4} \cos(10\pi x) \sin(6\pi \cdot 10^9 t - 54.4 z) \text{ (A/m)}$$

P. 6-21 $\bar{H}(x, z; t) = \bar{a}_y 2 \cos(15\pi x) \sin(6\pi 10^9 t - \beta z) \text{ (A/m)}$

Phasor with sine reference:

$$\bar{H} = \bar{a}_y 2 \cos(15\pi x) \cdot e^{-j\beta z} \quad (1)$$

$$\begin{aligned} \bar{E} &= \frac{1}{j\omega\epsilon_0} \nabla \times \bar{H} \\ &= \frac{1}{j\omega\epsilon_0} 2 \left[\bar{a}_x j\beta \cos(15\pi x) e^{-j\beta z} - \bar{a}_z 15\pi \sin(15\pi x) e^{-j\beta z} \right] \quad (2) \end{aligned}$$

$$\begin{aligned} \bar{H} &= -\frac{1}{j\omega\mu_0} \nabla \times \bar{E} \\ &= \frac{2}{\omega^2 \mu_0 \epsilon_0} \left[\bar{a}_y \left(-\frac{\partial E_z}{\partial x} + \frac{\partial E_x}{\partial z} \right) \right] \\ &= \bar{a}_y \frac{2}{\omega^2 \mu_0 \epsilon_0} \left[(15\pi)^2 + \beta^2 \right] \cos(15\pi x) \cdot e^{-j\beta z} \quad (3) \end{aligned}$$

Comparing (1) and (3), we require

$$\begin{aligned} (15\pi)^2 + \beta^2 &= \omega^2 \mu_0 \epsilon_0 = \frac{(6\pi 10^9)^2}{c^2} \\ &= \frac{(6\pi 10^9)^2}{(3 \times 10^8)^2} = 400\pi^2 \end{aligned}$$

$$\longrightarrow \beta = 13.2\pi = 41.6 \text{ (rad/m)}.$$

From (2), we have

$$\begin{aligned} \bar{E}(x, z; t) &= \text{Im}(\bar{E} e^{j\omega t}) \\ &= \bar{a}_x 496 \cos(15\pi x) \sin(6\pi 10^9 t - 41.6z) \\ &\quad + \bar{a}_z 565 \sin(15\pi x) \cos(6\pi 10^9 t - 41.6z) \text{ (V/m)}. \end{aligned}$$

Chapter 7

Plane Electromagnetic Waves

P. 7-1 a) In a source-free conducting medium with constitutive parameters ϵ , μ , and σ ,

$$\text{Eq. (7-62): } \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}. \quad (1)$$

$$\begin{aligned} \text{Eqs. (5-16a) } \nabla \times \nabla \times \vec{E} &= \nabla \nabla \cdot \vec{E} - \nabla^2 \vec{E} \\ \text{\&(7-61):} &= -\mu \frac{\partial}{\partial t} (\nabla \times \vec{H}). \end{aligned} \quad (2)$$

Substituting (1) in (2) and noting that $\nabla \cdot \vec{E} = 0$, we obtain the wave equation in dissipative media:

$$\nabla^2 \vec{E} - \mu \sigma \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0. \quad (3)$$

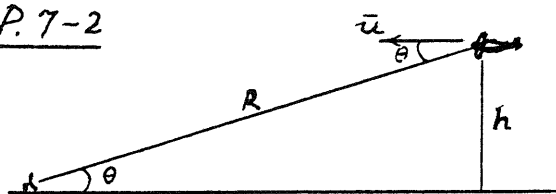
Similarly for $\vec{H}$.

b) For time-harmonic fields: $\frac{\partial}{\partial t} \rightarrow (j\omega)$ and $\frac{\partial^2}{\partial t^2} \rightarrow (-\omega^2)$.

Wave equation (3) converts to Helmholtz's equation:

$$\nabla^2 \vec{E} - j\omega\mu\sigma \vec{E} + k^2 \vec{E} = 0, \text{ where } k = \omega\sqrt{\mu\epsilon}.$$

P. 7-2



$$\Delta t = \frac{2R}{c} = 0.3 \times 10^{-3} \text{ (s)}.$$

$$\begin{aligned} R &= \frac{\Delta t}{2} c = \frac{0.3 \times 10^{-3}}{2} \times 3 \times 10^8 \\ &= 45 \times 10^3 \text{ (m)}, \\ &\text{or } 45 \text{ (km)}. \end{aligned}$$

$$h = R \sin \theta = 45 \times 10^3 \sin 15.5^\circ = 12 \times 10^3 \text{ (m)}, \text{ or } 12 \text{ (km)}.$$

$$\Delta f = 2f \left(\frac{u}{c} \right) \cos 15.5^\circ.$$

$$\rightarrow u = \frac{c \Delta f}{2f \cos 15.5^\circ} = 410.8 \text{ (m/s)}, \text{ or about } 1.2 \text{ Mach}.$$

P. 7-3 Assume that $\vec{H}(\vec{R})$ has the form:

$$\vec{H}(\vec{R}) = \vec{H}_0 e^{-jk \vec{a}_k \cdot \vec{R}}.$$

Then, From Eq. (6-80b),

$$\begin{aligned}\vec{E}(\vec{R}) &= \frac{1}{j\omega\epsilon} \nabla \times \vec{H}(\vec{R}) \\ &= \frac{1}{j\omega\epsilon} (-jk) \vec{a}_k \times \vec{H}(\vec{R}) \\ &= -\frac{1}{\omega\epsilon} (\omega\sqrt{\mu\epsilon}) \vec{a}_k \times \vec{H}(\vec{R}),\end{aligned}$$

or,
$$\vec{E}(\vec{R}) = -\eta \vec{a}_k \times \vec{H}(\vec{R}).$$

P. 7-4 $\vec{H} = \vec{a}_z 4 \times 10^{-6} \cos(10^7 \pi t - k_0 y + \frac{\pi}{4})$ (A/m).

a) $k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \frac{10^7 \pi}{3 \times 10^8} = \frac{\pi}{30} = 0.105$ (rad/m).

$\lambda = 2\pi/k_0 = 60$ (m).

At $t = 3 \times 10^{-3}$ (s), we require the argument of cosine in $\vec{H}$:

$$10^7 \pi (3 \times 10^{-3}) - \frac{\pi}{30} y + \frac{\pi}{4} = \pm n\pi + \frac{\pi}{2}, \quad n=0, 1, 2, \dots$$

$$\longrightarrow y = \pm 30n - 7.5 \text{ (m)} = 22.5 \pm n\lambda/2 \text{ (m)}.$$

b) Use phasors with cosine reference:

$$\vec{H} = \vec{a}_z 4 \times 10^{-6} e^{j(-k_0 y + \pi/4)} \quad (\text{A/m}).$$

From the result of Problem P. 7-3,

$$\begin{aligned}\vec{E} &= -\eta_0 \vec{a}_y \times \vec{a}_z 4 \times 10^{-6} e^{j(-k_0 y + \pi/4)} \\ &= -\vec{a}_x 4 \times 10^{-6} \eta_0 e^{j(-0.105 y + \pi/4)} \\ &= -\vec{a}_x 1.51 \times 10^{-3} e^{j(-0.105 y + \pi/4)} \quad (\text{V/m}).\end{aligned}$$

The instantaneous expression for $\vec{E}$ is:

$$\vec{E}(y, t) = -\vec{a}_x 1.51 \cos(10^7 \pi t - 0.105 y + \pi/4) \quad (\text{mV/m}).$$

P. 7-5 Use phasors with cosine reference.

$$\bar{E}(z) = \bar{a}_x 2 e^{-jz/\sqrt{3}} + \bar{a}_y j e^{-jz/\sqrt{3}} \quad (\text{V/m}).$$

a) $\omega = 10^8 \text{ (rad/s)} \longrightarrow f = 10^8/2\pi = 1.59 \times 10^7 \text{ (Hz)},$

$\beta = 1/\sqrt{3} \text{ (rad/m)} \longrightarrow \lambda = 2\pi/\beta = 2\sqrt{3}\pi \text{ (m)}.$

b) $u = \frac{c}{\sqrt{\epsilon_r}} = \frac{\omega}{\beta} \longrightarrow \epsilon_r = \left(\frac{\beta c}{\omega}\right)^2 = 3.$

c) Left-hand elliptically polarized.

d) $\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{\epsilon_r}} = \frac{120\pi}{\sqrt{3}} \quad (\Omega),$

$$\bar{H} = \frac{1}{\eta} \bar{a}_z \times \bar{E} = \frac{\sqrt{3}}{120\pi} (\bar{a}_y 2 e^{-jz/\sqrt{3}} - \bar{a}_x j e^{-jz/\sqrt{3}}),$$

$$\bar{H}(z,t) = \frac{\sqrt{3}}{120\pi} [\bar{a}_x \sin(10^8 t - z/\sqrt{3}) + \bar{a}_y 2 \cos(10^8 t - z/\sqrt{3})] \quad (\text{A/m}).$$

P. 7-6 Let $\alpha = \omega t - kz.$

$$\bar{E} = \bar{a}_x E_{10} \sin \alpha + \bar{a}_y E_{20} \sin(\alpha + \psi)$$

$$= \bar{a}_x E_x + \bar{a}_y E_y.$$

$$\frac{E_x}{E_{10}} = \sin \alpha, \quad \frac{E_y}{E_{20}} = \sin(\alpha + \psi)$$

$$= \sin \alpha \cos \psi + \cos \alpha \sin \psi$$

$$= \frac{E_x}{E_{10}} \cos \psi + \sqrt{1 - \left(\frac{E_x}{E_{10}}\right)^2} \sin \psi.$$

$$\left(\frac{E_y}{E_{20}} - \frac{E_x}{E_{10}} \cos \psi\right)^2 = \left[1 - \left(\frac{E_x}{E_{10}}\right)^2\right] \sin^2 \psi.$$

Rearranging:

$$\left(\frac{E_y}{E_{20} \sin \psi}\right)^2 + \left(\frac{E_x}{E_{10} \sin \psi}\right)^2 - 2 \frac{E_x E_y \cos \psi}{E_{10} E_{20} \sin^2 \psi} = 1,$$

which is the equation of an ellipse in E_x - E_y plane.

P.7-7 Given: $f = 3 \times 10^9$ (Hz), $\epsilon_r = 2.5$,

$$\tan \delta_c = \frac{\epsilon''}{\epsilon'} = 0.05.$$

a) Eq. (7-47): $\alpha = \frac{\omega \epsilon'' \sqrt{\mu}}{2 \sqrt{\epsilon'}} = \frac{\omega}{2} \left(\frac{\epsilon''}{\epsilon'} \right) \frac{\sqrt{\epsilon_r}}{c} = 2.48$ (Np/m).

$$e^{-\alpha x} = \frac{1}{2} \rightarrow x = \frac{1}{\alpha} \ln 2 = 0.279 \text{ (m)}.$$

b) Eq. (7-49): $\eta_c = \frac{1}{\sqrt{\epsilon_r}} \sqrt{\frac{\mu_0}{\epsilon_0}} \left(1 + j \frac{\epsilon''}{2\epsilon'} \right) = \frac{238}{\sqrt{2.5}} \angle 1.43^\circ \text{ } (\Omega)$
 $= 238 / 0.008 \pi \text{ } (\Omega).$

Eq. (7-48): $\beta = \omega \sqrt{\mu \epsilon} \left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right] = 31.6 \pi$ (rad/m).

$$\lambda = \frac{2\pi}{\beta} = 0.063 \text{ (m)}.$$

$$u_p = \frac{\omega}{\beta} = 1.897 \times 10^8 \text{ (m/s)}.$$

$$u_g = \frac{1}{\frac{d\beta}{d\omega}} = \frac{c}{\sqrt{\epsilon_r}} \left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right] = 1.898 \times 10^8 \text{ (m/s)}.$$

c) At $x=0$, $\bar{E} = \bar{a}_y e^{j\pi/3}$, $\bar{H} = \frac{1}{\eta_c} \bar{a}_x \times \bar{E} = \bar{a}_x 0.210 e^{j(\frac{\pi}{3} - 0.008\pi)}$

$$\bar{H}(x,t) = \bar{a}_x 0.210 e^{-2.48x} \sin(6\pi 10^9 t - 31.6\pi x + 0.325\pi) \text{ (A/m)}.$$

P.7-8 Both copper and brass are good conductors,

$$\left(\frac{\sigma}{\omega \epsilon} \right)^2 \gg 1: \alpha = \sqrt{\pi f \mu \sigma}, \delta = \frac{1}{\alpha}, \eta_c = (1+j) \frac{\alpha}{\sigma}.$$

a) $f = 1$ (MHz).

	$\eta_c (\Omega)$	α (Np/m)	α (dB/m)	δ (m)
Copper	$2.61(1+j) \times 10^{-4}$	1.51×10^4	1.31×10^5	6.61×10^{-5}
Brass	$4.98(1+j) \times 10^{-4}$	0.79×10^4	0.69×10^5	12.6×10^{-5}

b) $f = 1$ (GHz).

	$\eta_c (\Omega)$	α (Np/m)	α (dB/m)	δ (m)
Copper	$8.25(1+j) \times 10^{-3}$	4.79×10^5	4.16×10^6	2.09×10^{-6}
Brass	$1.58(1+j) \times 10^{-3}$	2.51×10^5	2.18×10^6	3.99×10^{-6}